

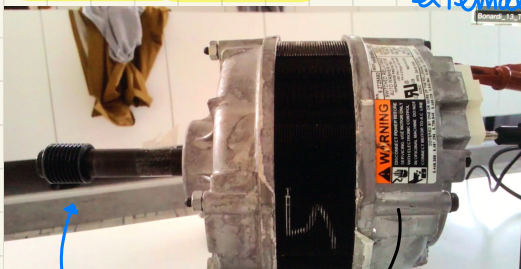
# SWITCHED RELUCTANCE MACHINE

[24/05]

# SWITCHED RELUCTANCE MACHINE

## machine structure

externally similar to the other machines studied:



ROTATING shaft

strange machine,  
NOT easy to buy on the market..

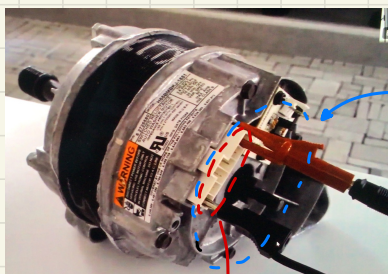
(used in washing machine)

+ back mechanical sensors  
LED/phototransistor

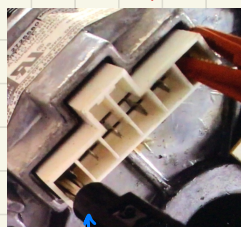
↓  
mech sensor to find

actual position of the ROTOR

(encoder!) → disk with holes segments



terminals



6 terminals

here we have strange connection..

(4 COILS) ↓

middle terminals

## EXPERIMENTS..

### • DISCONNECTED MACHINE:

try to move ROTOR → low damped system, move very well (low friction)

### • SHORT CIRCUIT machine: still ROTOR move well even if short circuit

2 coils terminals → NO effect! means that there is **no permanent magnet**

[sometimes on washing machine we find Induction machine → made by ROTOR coils / squirrel cage] --> 3 phase connection on ROTOR

• IF apply voltage on the machine, connect terminals to P.S (you hear noise.) → there is movement + oscillation

- And when it is supplied: you are NOT able to move it by hand!  
(you can move, but it makes opposite torque  $\rightarrow$  Resistive)

$\Downarrow$   
and ROTATING by hand

$\downarrow$   
LOW COST (NODM)

simple machine  $\rightarrow$  you can choose the ROTOR position  
applying a constant voltage (current)



you can identify overall  
8 equilibrium position  
(stable position)

{ different from IM } { DC current on IM there is friction due }  
which has a slip! { to eddy current on ROTOR.. }

$\downarrow$   
similar effect here, some friction when supplied

$\Downarrow$   
**machine model THEORY:**

(different structure from the other machines)

Energy balance a lot different..

called: "six for"

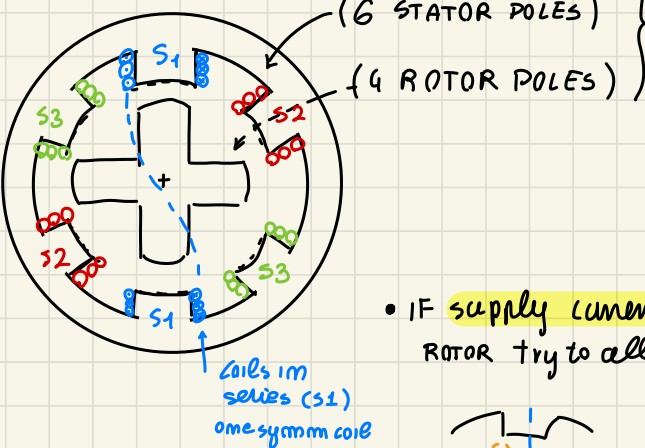
**6 4 RELUCTANCE machine**

$\downarrow$   
 **$N_s$ : stator poles number (6)**

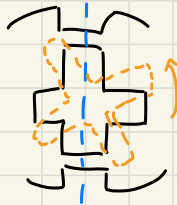
**$N_r$ : rotor poles number (4)**

**$q$ : number of phases (3)**

[ANISOTROPIC machine]



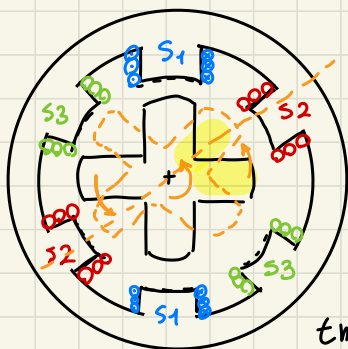
- IF supply current on S1  
ROTOR try to align



and IF I try to move  
the ROTOR  $\rightarrow$  there is a  
Torque which try to move  
rotor position again on  
initial configuration...

→ to move the ROTOR CCW :

the new phase I need to supply after  $s_1$



↓  
supply  $S_2$ , so ROTOR try to align  
with  $S_2$  pole pair → move CCW  
(If I supply  $S_3$  otherwise It move CW  
try to align to nearest pole pair!)

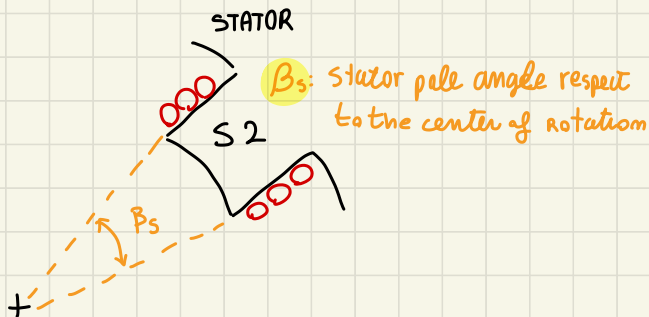
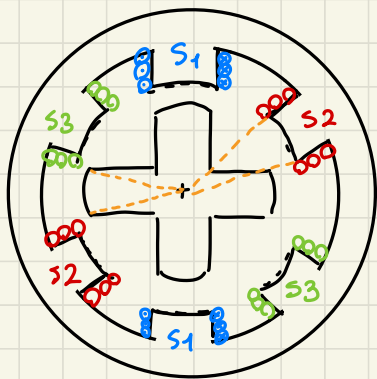
$S_2$  applied try to align nearest  
ROTOR pole to it

↓  
this generated torque for alignment is called  
**RELUCTANCE torque**

↓  
the usual way to supply this machine is by 1 phase each time  
(only 1 phase)

supply:  $S_1 \rightarrow S_2 \rightarrow S_3 \dots$  moving CCW  
while  $S_1 \rightarrow S_3 \rightarrow S_2$  move CW

↓ to study that machine we need to introduce other elements :



$B_r$  angle of ROTOR size respect center

↳ machine dyn model ⇒

model classical coil model:

$$V_{s1} = R_s i_{s1} + p \psi_{s1}$$

(true for all machine)  
except for DC machine  
with commutator

because it works with one

phase active each time, all other currents are equal to 0  
(only  $i_{s2} \neq 0$ )

$$\psi_{s1} = L_{s1} i_{s1}$$

constant? OR it changes with  $\theta_m$  (ROTOR position)

(consider ROTOR on that angular position)

where  $L_{s1}$  is maximum?  
(max self inductance?)

FOR  $\theta_m = 0$

or  $\theta_m = \pi/2$

$\theta_m = \pi$

$\theta_m = 3/2\pi$

max  
 $L_{s1}$

(mech angle reference)

for this  $\theta_m$   
flux lines induced by  $i_{s2}$   
pass through min air gap  $\rightarrow$  max inductance  
(high permeance, high inductance!)

$$\theta_m = \pi/4$$

While the minimum

$$L_{s1} \text{ for } \theta_m = \pi/4$$

MIN  $L_{s1}$

MAX  $L_{s1}$

small air gap

flux lines given by  $i_{s2}$

$\theta_m = 0$   
(and  $\pi/2, \pi, 3/2\pi$   
same ROTOR  
position...)

$\Rightarrow$  so  $L_{s1}$  is function of mech angle  
(similarly to IPM, synchrel ... where there  
is ROTOR anisotropy)

$\hookleftarrow$  Here I have ANISOTROPY from

ROTOR looking to STATOR

and STATOR looking to ROTOR  $\rightarrow$  but no coils on ROTOR (no problem)!

Anisotropy  $\rightarrow L_{s1}(\theta_m)$

from **ENERGY BALANCE**: it gives you the following idea:

$$\underbrace{V_{s1} i_{s1}}_{\text{total power inside machine}} = \underbrace{R_{s1} i_{s1}^2}_{\text{Joule losses}} + \underbrace{i_{s1} p \psi_{s1}}_{\text{What is this term?}}$$

What is this term?

usually we have another machines  $dU/dt$ , but we had computed previously  $dU/dt$  when  $L_s = \text{constant}$ !

while here  $L_s(\theta_m)$  is function of  $\theta_m$  NOT constant

(also for IPM, we move to a Ref. fixed to

ROTOR: obtaining a viewpoint in which  $L_d, L_q$  constant)

until now we obtain representation with constant  $L \Rightarrow$  HERE we have **variable inductance** to manage!

$\Downarrow$  compute  $i_{s1} p \psi_{s1}$ : (now in 3 terms)

$$V_{s1} i_{s1} = R_{s1} i_{s1}^2 + \underbrace{i_{s1} p L_{s1}(\theta_m) i_{s1}}_{\text{which is contribution (1/2 Pm, 1/2 U) related to Pm, and which one related to variation of energy stored into inductances?}} + \underbrace{i_{s1} \frac{d i_{s1}}{dt} \cdot L_{s1}(\theta_m)}_{\text{similar with 1/2 term of difference!}}$$

IF we directly compute derivative of energy stored into inductance:

$$U = \frac{1}{2} L_{s1}(\theta_m) \cdot i_{s1}^2 \rightarrow \frac{dU}{dt} = \underbrace{\frac{1}{2} \frac{dL_{s1}(\theta_m)}{dt} i_{s1}^2}_{\text{related to Pm}} + \underbrace{L_{s1}(\theta_m) i_{s1} \frac{d i_{s1}}{dt}}_{\text{related to } dU/dt}$$

$\downarrow$

so the term  $V_{s1} i_{s1} \rightarrow p L_{s1}(\theta_m) i_{s1}^2 = \underbrace{\frac{1}{2} p L_{s1}(\theta_m) i_{s1}^2}_{\text{related to Pm}} + \underbrace{\frac{1}{2} p L_{s1}(\theta_m) i_{s1}^2}_{\text{related to } dU/dt}$

so we obtain

$$P_m = \frac{1}{2} \frac{dL_{s1}(\theta_m)}{dt} \cdot i_{s1}^2$$

LOT different from the other machines!

(similar on DC brushless that we study as isotropic! otherwise you need also  $dL/dt$  on DC brushless: reluctance torque simplified)

so you can manage  $P_m = \frac{1}{2} \frac{dL_{s1}(\theta_m)}{dt} i_{s1}^2$

with  $L_{s1}(\theta_m)$   
function of mech angle  $\theta_m$   
which is  $\theta_m(t)$

calculate it as: ↓

$$P_m = \frac{1}{2} \frac{dL_{s1}(\theta_m)}{d\theta_m} \left( \frac{d\theta_m}{dt} \right) i_{s1}^2$$

( $m_p = 1$  IF we supply only one STATOR... coil)

$m_p \Omega_m$  (With  $m_p$  HARD to define on this machine) ↓

but anyway we relate elect/mech world by  $m_p$  (# pole pairs)

$$T_e = \frac{P_m}{\Omega_m} = \frac{1}{2} m_p \frac{dL_{s1}(\theta_m)}{d\theta_m} i_{s1}^2$$

SO: IF you supply this machine from a stable position, IF change the sign on some terminal NOTHING change on  $T_e \rightarrow T_e$  is function of  $i_{s1}^2$  so does NOT depends on current sign!  
( $i_{s1} \geq 0$  nothing change if same  $|i_{s1}|$ )

( $T_e = 0$  in some position..)

$T_e$  depends on  $|i_{s1}| \rightarrow$  you don't need  $i_{s1} \geq 0$  on P.S, it reduces the complexity of power supply!

NOW, the problem is to FIND  $L_{s1}(\theta_m)$ ?

we need to study ↓

the structure and the relative movement between ROTOR and STATOR

↓

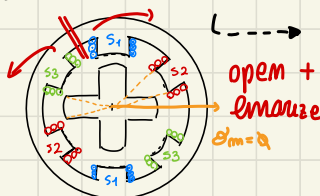
consider  $\beta_s < \beta_r$  (then we will study for  $\beta_r > \beta_s$  and find same shape..)

↓

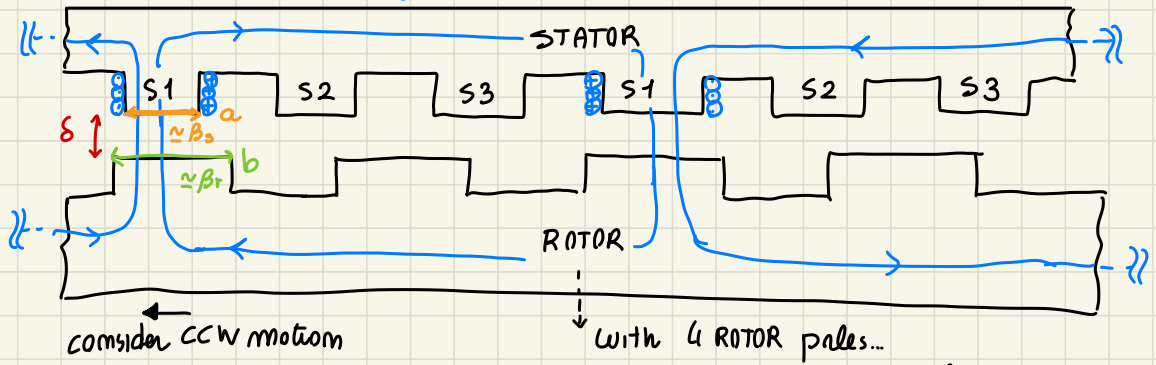
usually we have  $N_r < N_s$  (so it make sense to have  $\beta_s < \beta_r$  to have more space)  
(DESIGNER cue DOF on that choice)

• START from  $\theta_m = 0$

let's study the machine by "opening" it, and "linearize" the struine



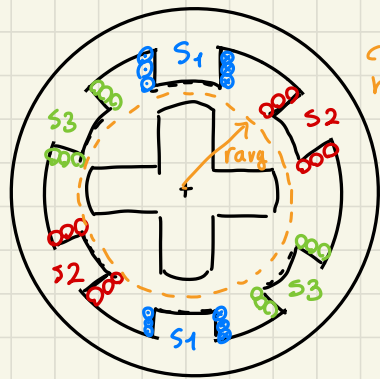
(Flux lines of  $i_{s1}$ )



that length  $a$  is equal to  $\beta_s$  times the radius

considering  $\delta \ll 1$  (small air gap)

→ Radius of ROTOR poles  $\approx$  average value of radius inside air gap  
(true also for  $\beta_s$ )



consider  
avg inside  
air gap as  
common  
radius of  
ROTOR and  
STATOR

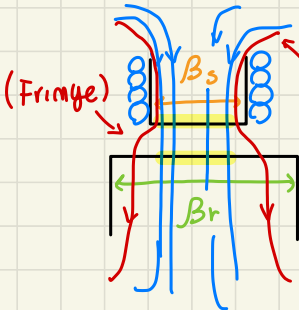
→ so I can consider the  
length  $a, b$

$\{a \sim \beta_s\} \Rightarrow$  I don't consider  
the RADIUS, I can  
simply take  
the angle, NOT  
the radius!

simplify  
representation

↓  $\mathcal{G}_m = \mathcal{G}$ :

looking to one S1 pole of STATOR



I NEGLECT  
those lines which  
try to move from  
STATOR to ROTOR outside  
stator... NO FRINGE EFFECT

(small effect, I consider so same  
cross section area for STATOR / ROTOR)

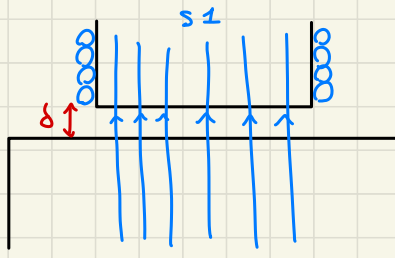


Neglect  
fringe!

⇒

so overall emf as:

$\delta$ : air gap



[ Reluctance of this magnetic circuit? ]

easy to compute

( $\mu_{Fe} \text{ RELUCTANCE} \approx 0$ )  $\mu_{Fe} \gg \mu_0$

$\mathcal{G}_{Fe} \approx \infty$  (high permeability  $Fe$ )

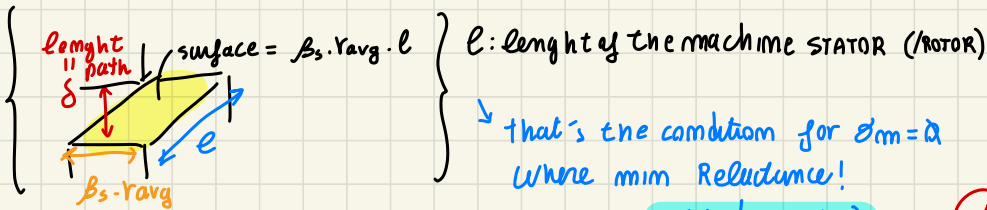
consider  $\mu_{Fe} \approx +\infty$

we consider only  $\mathcal{G}_0$  (air gap reluctance)

$$\mathcal{G}_0 = \frac{1}{\mu_0 \beta_s \cdot \text{avg} \cdot l} \delta$$

(related to one pole of  $S_1$ )

↳ the overall  $\mathcal{G}_0$  is twice this!



↳ that's the condition for  $\mathcal{G}_{\text{min}} = \infty$  where min Reluctance!

⇒ so max  $L_{s2}(\theta_m)$

"Regions"

(1) of  $L_{s2}(\theta_m)$

[ this condition is maintained also for small rotor motion! ]

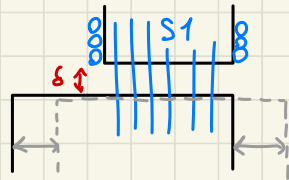
↳ In which  $\theta_m$  angle range we maintain max  $L_{s2}$ ?

that's the last point on which  $L_{s2}$  has its max value..

corresponding to

$$\theta_m = \frac{\beta_r - \beta_s}{2}$$

(last point where  $L_{s2}(\theta_m)$  self inductance remain max value)

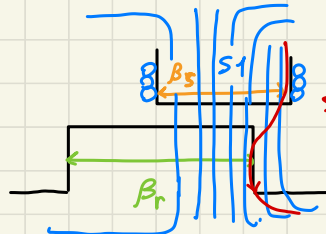


moving the ROTOR by  $\left( \frac{\beta_r - \beta_s}{2} \right)$

still all STATOR is flux lines goes through SMALL AIR GAP!

continuing to move  $\theta_m > \frac{\beta_r - \beta_s}{2}$

(2)

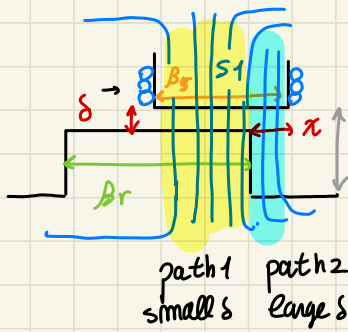


still neglect fringe effect

lines tends to curve to ROTOR body!

(not had approx!)

$$\sigma_m > \frac{\beta r - \beta_s}{2}$$



call this length  $x$

$\Rightarrow$  equivalent magnetic circuit?

FLUX pass through 2 path  
2 parallel flux path

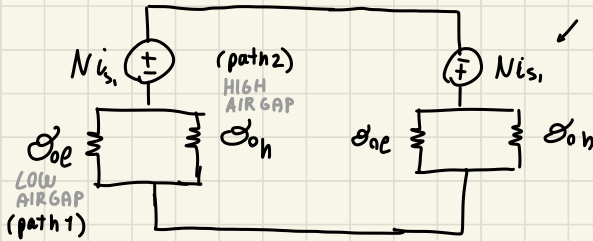
equivalent magm circuit as two parallel reluctance

(call overall distance  $\delta_h$  and  $\delta$ )

in this case...

considering  $\sigma_{fe} \approx \infty$

( $\mu_{fe} \approx +\infty$  respect  $\mu_0$ )  $\Rightarrow$  so we consider only air reluctance  $\frac{1}{2} \sigma_{fe} \approx$



from the 2 poles of  $i_s$  we have 2 Magnetomotive force

$$\begin{cases} \sigma_{oe} = \frac{1}{\mu_0} \frac{\delta}{(\beta_s - x) r_{avg} \cdot l} & \text{(small air gap reluctance, reduced by } x \text{ at den)} \\ \sigma_{oh} = \frac{1}{\mu_0} \frac{\delta_h}{x r_{avg} \cdot l} \end{cases} \Rightarrow \lambda_{oe} = \frac{1}{\sigma_{oe}} \quad \lambda_{oh} = \frac{1}{\sigma_{oh}}$$

to find easily the parallel (simpler computation)

as total parallel permeance  $\rightarrow$  as on electric circuit parallel conductance  $g_1 // g_2$  equivalent to  $g_1 + g_2$

$$\lambda_{TOT} = \lambda_{oe} + \lambda_{oh} = \mu_0 r_{avg} \cdot l \left[ \frac{\beta_s - x}{\delta} + \frac{x}{\delta_h} \right]$$

while  $x_{MAX}$  value:

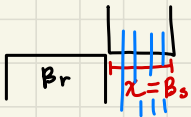
(When all lines go through

$\delta_h$ )  $\rightarrow$  when  $x = \beta_s$  min  $\lambda$

so min  $L \dots \Rightarrow$  contribute of small air gap  $\delta$

goes to  $\infty \rightarrow$  all given by  $\left( \frac{x}{\delta_h} \right) \rightarrow \lambda_{TOT}$  LOWER!

$\uparrow$  FOR  $x = \beta_s$   $\delta_h$  don't contribute on  $\lambda_{TOT}$  (AS BEFORE!)



so after  $B_s$  from  $\frac{B_r - B_s}{2}$  we have  $\min \lambda \rightarrow$  so  $\min$  inductance  $L_{s1}(\delta_m)$

and  $\lambda_{TOT}$  decreases linear with  $x$

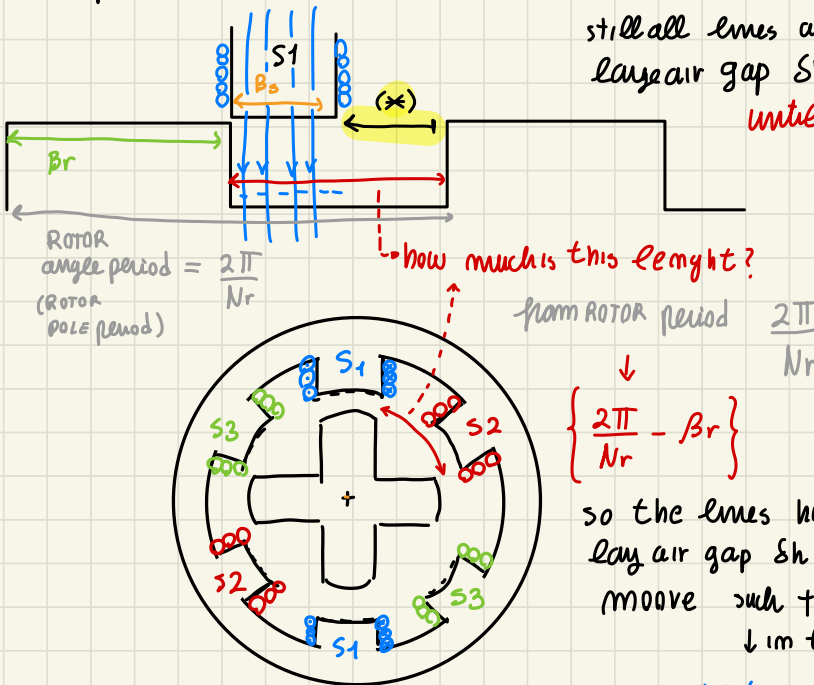
(so linear with  $\delta_m$ ) inside  $\left( \frac{B_r - B_s}{2}, \frac{B_r - B_s}{2} + B_s \right)$

$\downarrow$   
so because  $L = N^2 \lambda_{TOT}$  linearly dependent on displacement  $x$   
also inductance decreases linearly!

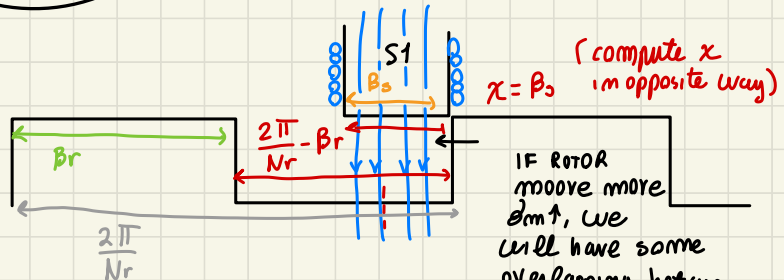
BUT there is another condition to analyze on  $L_s(\delta_m)$

shape:

IF ROTOR moves more  $\delta_m \uparrow$   
still all lines as to go through  
large air gap  $\delta_h$  ( $\min L_{s1}$ )  
until..



so the lines has to pass through  
large air gap  $\delta_h$  until ROTOR has  
move such that  
 $\downarrow$  in this configuration



IF ROTOR  
move more  
 $\delta_m \uparrow$ , we  
will have some  
overlapping between  
ROTOR/STATOR, so lines  
going through small  
air gap  $\delta$  path

to reach this  
new position need  
a displacement of  $(x)$

$$\frac{2\pi}{N_r} - B_r - \beta_s$$

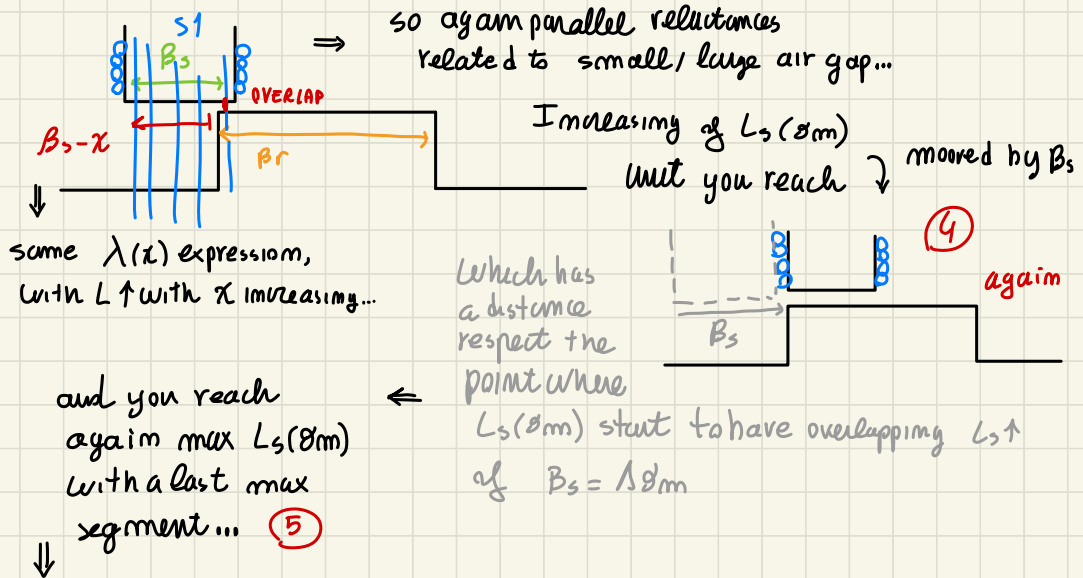
and we maintain  $L_s(\theta_m) = L_{s\min}$  constant for

$$\theta_m \in \left( \frac{\beta_r - \beta_s}{2} + \beta_s, \frac{\beta_r - \beta_s}{2} + \beta_s + \left( \frac{2\pi}{N_r} - \beta_r - \beta_s \right) \right)$$

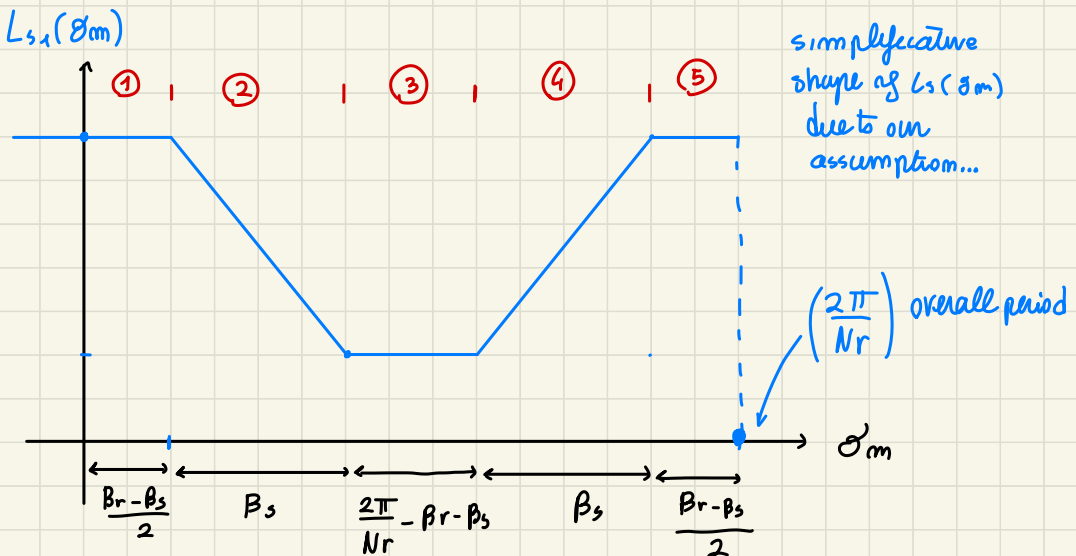
then finally,

after that point we

have ROTOR / STATOR pole overlapping

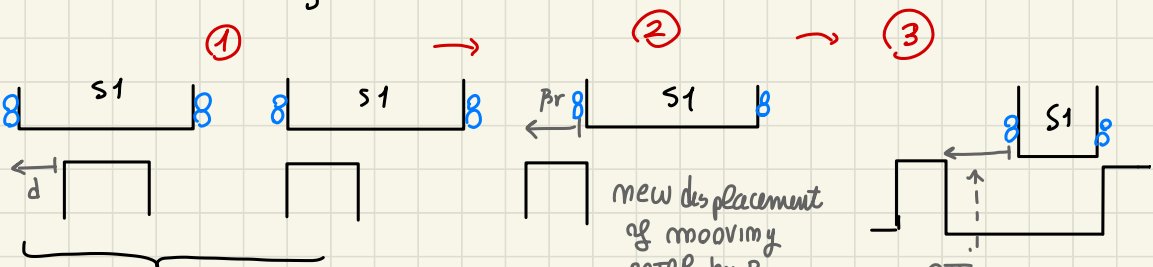


SO OVERALL you can plot  $L_{s1}(\theta_m)$  on that studied regions



overall period  $\frac{2\pi}{N_r} \rightarrow$  then you have another ROTOR poles, you START again for  $N_r$  times with same shape of  $L_{s1}(\theta_m)$

NOW... considering the case  $\beta_s > \beta_r$  design choice  
the condition to study are:



max between that with overall displacement on 2 equal to  $d = \frac{\beta_s - \beta_r}{2}$  (opposite sign of previous displacement)

new displacement of moving ROTOR by  $\beta_r$   
 $\downarrow$   
so change on  $L_s(\theta_m)$  that range as

$\frac{2\pi}{N_r} - \beta_r - \beta_s$   
distance until you remain with min  $L_s$  (same as before!)

we put the module  $\left| \frac{\beta_r - \beta_s}{2} \right|$  to represent both conditions!

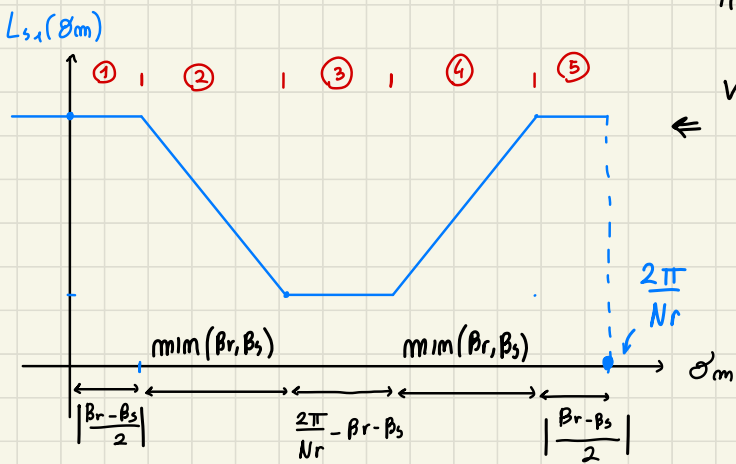
$\min(\beta_r, \beta_s)$  to represent both condition!

from previous to new condition where all lines go through small air gap

and then goes periodically..

until you move by  $\frac{\beta_s - \beta_r}{2}$

moving by  $\beta_r$  increase  $L_s$



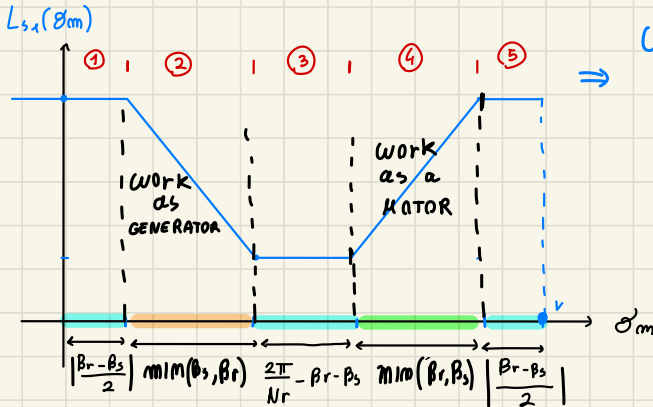
Valid both for  $\beta_s > \beta_r$  OR  $\beta_r > \beta_s$  designing machine..

## TORQUE

$$T_e = \frac{m_p}{2} \frac{dL_{s1}(\theta_m)}{d\theta_m} i_{s1}^2$$

## STRATEGY

of  $i_{s1}$ ?  $\Rightarrow$  how to control  $i_{s1}$  to maintain constant torque?



When  $L_{s1}(\theta_m) = \text{constant}$

$$\frac{dL_{s1}}{d\theta_m} = 0 \rightarrow T_e = 0 \text{ NO}$$

We have to choose between the pos/neg slopes of  $L_{s1}(\theta_m)$

because  $T_e$  is not influenced by  $\text{sign}(i_{s1})$

$T_e \geq 0$  according to  $dL_{s1}/d\theta_m$

$\Leftarrow$  so depends on slope of  $L_{s1}(\theta_m)$

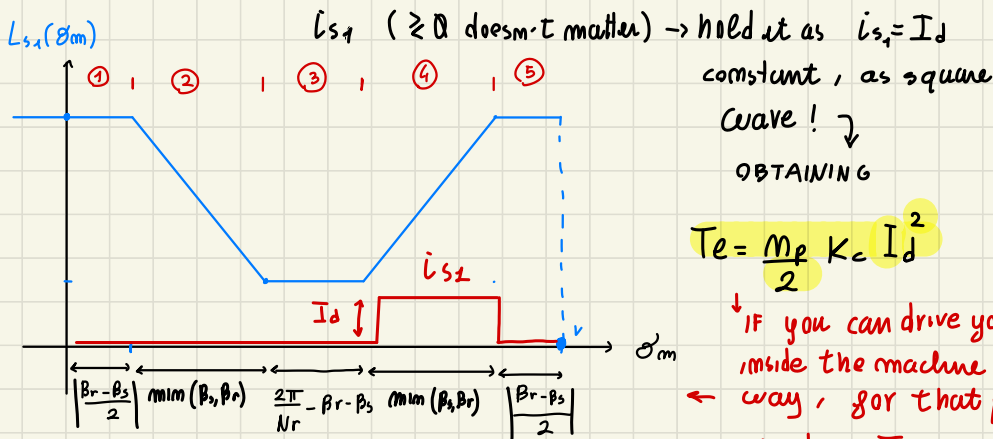
WORK as MOTOR:  $T_e > 0$

to work as GENERATOR:  $T_e < 0$

↑  
you can use this machine in both condition..

$\rightarrow$  so if ideally  $L_{s1}(\theta_m)$  has constant slope on that region.. we have  $T_e$  constant if  $i_{s1}$  is constant

IDEA: introduce a current constant

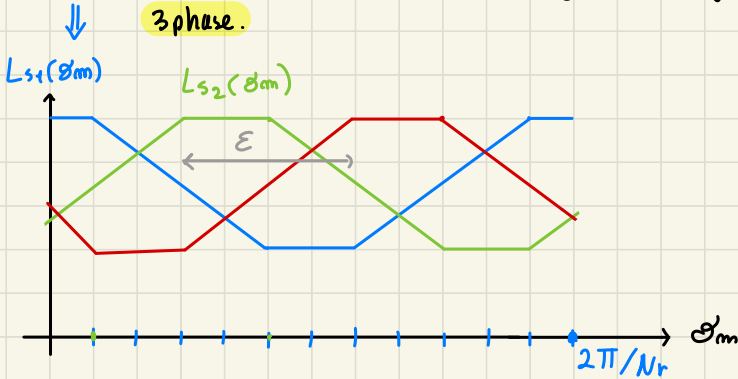


constant, as a square wave!  $\rightarrow$  OBTAINING

$$T_e = \frac{m_p}{2} K_c I_d^2$$

↓ if you can drive your current inside the machine in this way, for that period we have  $T_e$  constant

so, to have **Te constant OVERALL**  $\forall t$ ... you need to think about real machine STATOR structure, made out of 3 windings...



To obtain  $L_{s2}(\theta_m)$ ... after some time.. due to displacement you now 3phase, divide period  $\frac{2\pi}{N_r}$  in 2 part, considering that || ?  
We have 4 phases... (NOT 3)

↓  
there is some symmetry on  $L_s(\theta_m)$  → Fill the overall period by  $L_s$  and displacement  
inductance  
between this inductances is equal to  $\Rightarrow \frac{1}{3} \left( \frac{2\pi}{N_r} \right)$

$$\epsilon = \frac{2\pi}{N_r} \cdot \frac{1}{3} \quad (\text{in general!}) \quad \begin{array}{l} \text{displacement} \\ \text{between the inductances} \end{array}$$

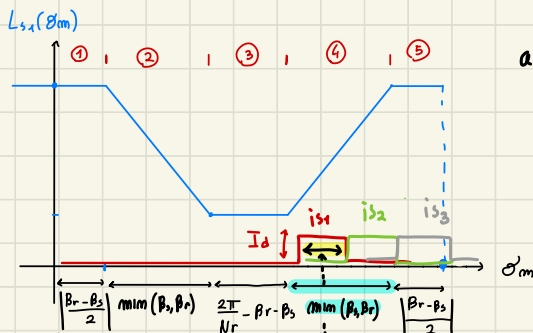
↓  
In our experimental machine, we had 8 stable  $T_e$  points...  
this means that:

( $N_r = 4 \rightarrow 2$  rotor pole pairs...) ↑ while in machine FROM ROTOR P.O.V you have 8 ROTOR poles  
machine

↓  
{ because you have constant max  $T_e$  only in some positions, only torque 4 times... }

to have a **CONSTANT TORQUE** → **STRATEGY?**

the constraints to maintain to guarantee  $T_e$  const?



as found before  $i_s$  is constant  
for a period equal to...

and repeat also for  $i_{s2}, i_{s3}$   
(have room for  $i_{s2}, i_{s3}$  maintained)

↓ so we divide the period into  
2 times for each phase

$\varepsilon$ : maintained

for  $\varepsilon = \frac{2\pi}{Nr \cdot q}$ , some displacement between  
these self inductances.

↓  
because after that point you start  
with next current...

to have  $T_e = \text{const}$  you  
need to satisfy that

$$\varepsilon = \frac{2\pi}{Nr \cdot q} \leq \min(\beta_s, \beta_r)$$

↓  
introduce  $i_s$   
constant with const slope...

constant

contribute for a period equal to  $\varepsilon$ ! →  $T_e$  constant

← Where you introduce  $i_s = I_d$   
you have constant positive  
 $T_e$  torque (so MOTOR)

+ other const limits (complex one, we will see it...)

# [29/05] CONTINUE, SWITCHED RELUCTANCE MACHINE (SRM)

## Dyn equations.

(I WINDING)

$$\begin{cases} V_{s1} = R_s i_{s1} + p \Psi_{s1} \rightarrow \text{and from energy balance:} \\ \Psi_{s1} = L_{s1}(\theta_{rm}) i_{s1} \end{cases}$$

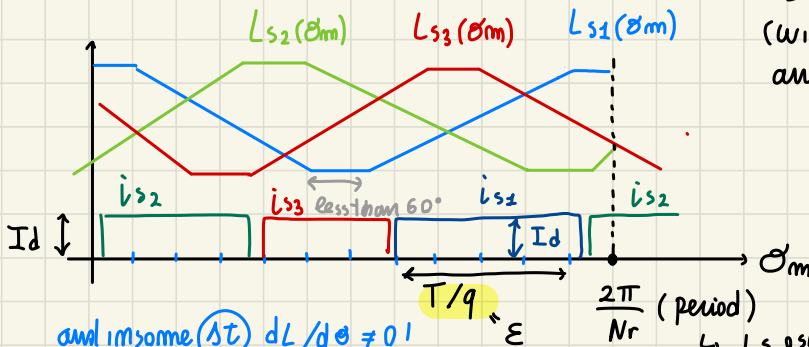
self inductance

$$T_e = \frac{m p}{2} K_c i_d^2$$

$$K_c = \frac{d L_{s1}(\theta_{rm})}{d \theta_{rm}}$$

and looking on relative position

between ROTOR/STATOR poles, we introduce: (simplification also made)



- 3 windings (with  $q = 3$  machine)
- and  $\begin{cases} N_s = 6 \\ N_r = 4 \end{cases}$

and in some case  $dL/d\theta \neq 0$ !

to have  $T_e$  we use this period, in particular, where  $dL/d\theta > 0$  ( $T_e > 0$ )  $\rightarrow$  MOTOR (no dependency on  $i_s$  sign)

(while where  $dL/d\theta < 0 \rightarrow T_e < 0$ )  
GENERATOR

usually machine used for high speed application!

(like washing machine or airplane turbine, to START the turbine (like motor) producing then energy to supply)

In order to obtain  $T_e \sim \text{constant}$  on motor, we supply  $i_{s1}$  properly with constant amplitude  $I_d$ , as square wave

similarly for  $i_{s2}$ , same for  $i_{s3} \rightarrow$  supplied on st where correspondingly  $L_{s1}$  has positive slope!

$\Downarrow$   
so the period on which you maintain a phase current constant is

$$T/q \text{ with } T = 2\pi/N_r \text{ called } \epsilon = \frac{T}{q} = \frac{2\pi}{N_r q}$$

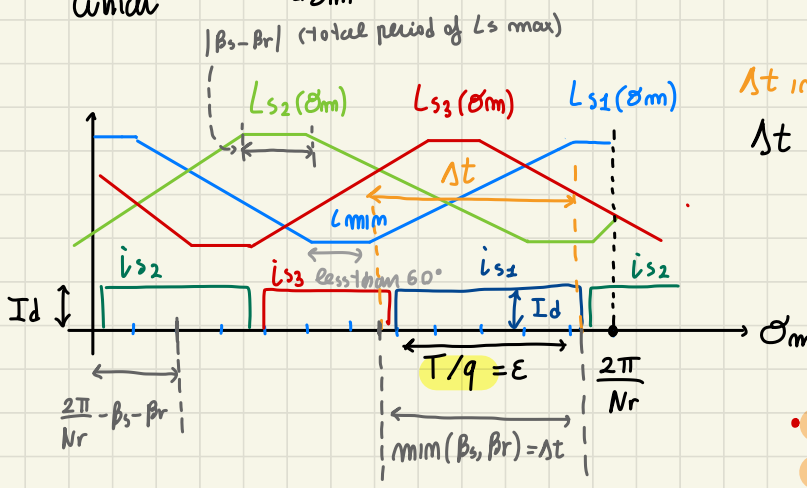
because on one period  $T$  you need to supply all your machine coils

⇒ for each  $\varepsilon$  torque is given by different  $i_{s_j}$   $j=1,2,3$  according to  $L_{s_j}$  with positive slope here (as on DC brushless..)

↓

to have  $T_e = \text{constant}$  you need to have:

period during which  $K_c = \frac{dL_{s_j}(\theta_m)}{d\theta_m} \neq 0 \geq$  to the current period...



$\Delta t$  in which slope positive

$\Delta t \geq$  to current period  $T/q = \varepsilon$

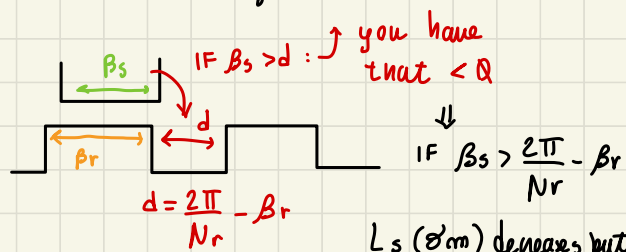
otherwise on some time we have  $T_e = 0$

↓

• (1<sup>st</sup>) constraint:  
 $\varepsilon \leq \min(\beta_s, \beta_r)$   
 $\Delta t$

• (2<sup>nd</sup>) constraint:

$\frac{2\pi}{Nr} - \beta_s - \beta_r \geq 0 \rightarrow$  this means that...  
 IF this is  $< 0$  you have



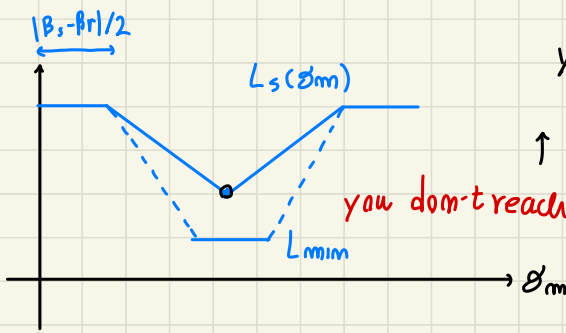
$L_s(\theta_m)$  decreases but don't reach  $L_{smin}$  never!

While when :

(2<sup>nd</sup> constraint) guaranteed

--- here you have  $L_{smin}$ ! (you reach min!)

$L_s$  if this constraint is violated... ⇒



you need

$$\frac{2\pi}{N_r} - \beta_s - \beta_r \geq 0$$

to reach properly  $L_{min}...$

↑

you don't reach  $L_{min}!$

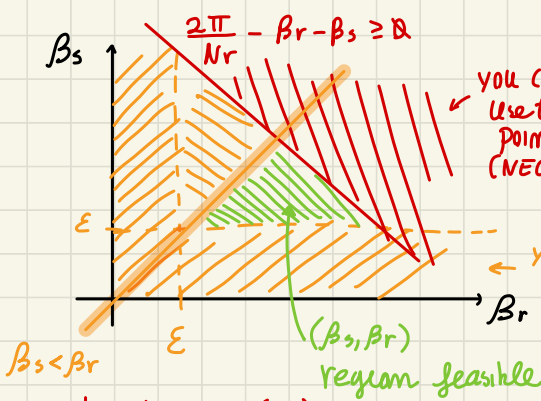
meaning also that you have a lower slope of the  $L_s(\delta_m)$ , IF you reach  $L_{min}$  you have high neg slope OK

so this constraint is good as design choice

so an higher torque!  
( $T_e \sim \frac{dL}{d\delta}$ )

↓

## DESIGN $(\beta_s, \beta_r)$ "OPERATING REGIONS" CONSTRAINTS



{ all feasible  $(\beta_r, \beta_s)$  }  
you can use to design your machine!

(I)

you cannot use this one, because it violate 1st...

having  $\beta_s < \epsilon$   
 $\beta_r < \epsilon$  }  $\min(\beta_s, \beta_r) < \epsilon$  !

2<sup>nd</sup> condition. (II)

$\frac{2\pi}{N_r} - \beta_s - \beta_r \geq 0$  linear constraint with neg slope of 45° deg

(free choice, but usually)

usually machine designed such that  $N_s > N_r \Rightarrow \beta_s < \beta_r$

↑

Design region to realize this machine

due to room you have inside machine

# CONTROL SCHEME

from dym eq:  $V_{s2} = R_s i_{s2} + \frac{d\psi_{s1}}{dt} = R_s i_{s2} + L_{s1}(\theta_{m}) \frac{di_{s1}}{dt} + i_{s2} \frac{dL_{s1}(\theta_{m})}{dt}$

$V_{s2} = \underbrace{R_s i_{s2}}_{\text{R}_s \text{ voltage drop}} + \underbrace{L_{s1}(\theta_{m}) \frac{di_{s1}}{dt}}_{\text{self inductance effect}} + \underbrace{i_{s2} \frac{dL_{s1}(\theta_{m})}{d\theta_{m}}}_{e \text{ (treated as "disturbance")}} M_p \Omega_m$

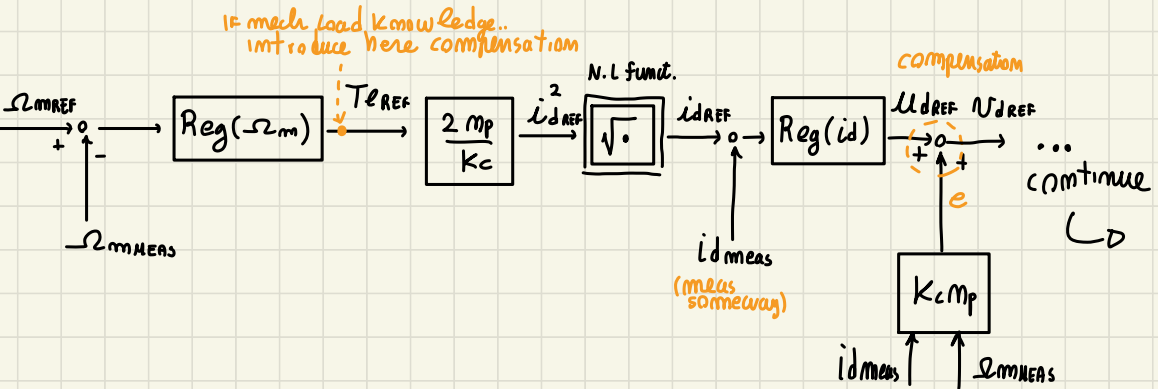
for the  
control  
scheme

$u_d$   
↑  
control  $u_d$  to  
force the current

↓ classical control scheme,  
with a speed ref compared  
with meas of  $\omega$ ..

Something proportional  
to the speed (like emf)  
↓  
(STRANGE DROP!)  $e_s$  represent NOT  
only mech power conversion  
( $P_{m} \neq \text{emf} \cdot i_s$ ) → this include also  
part of  $dV/dt$ ... NOT only power  
related to  $P_m$

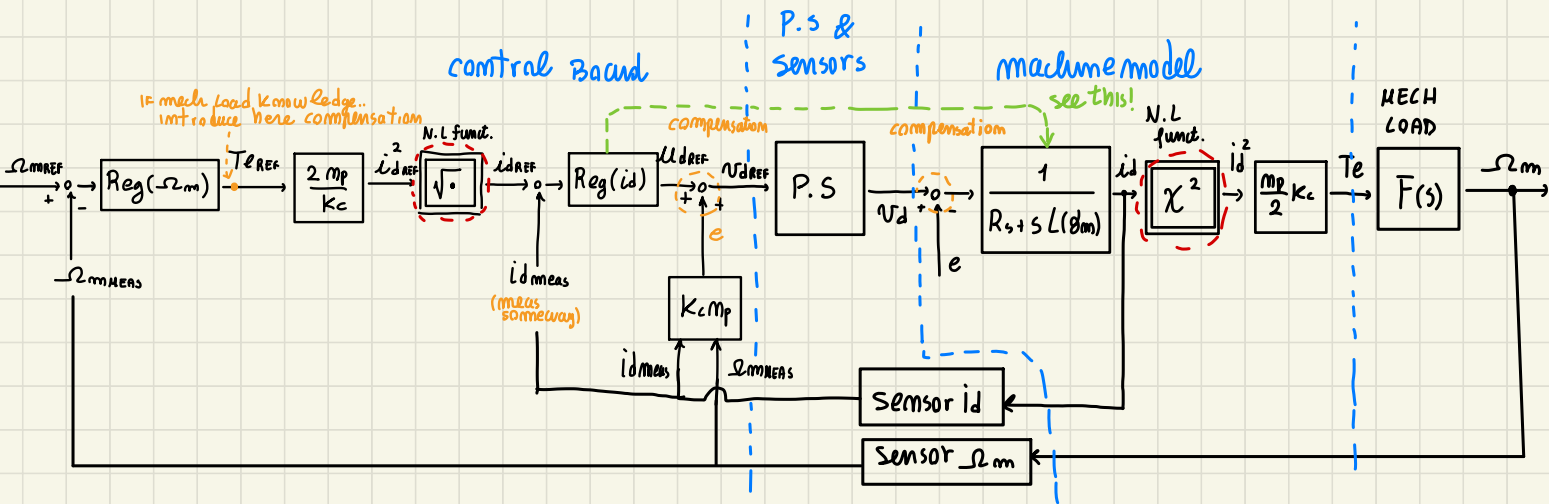
$$T_e = \frac{m_p}{2} K_c i_d^2$$



Next page..

## Control scheme

$$\begin{cases} v_{s\pm} = R_s i_{s\pm} + L_{s\pm}(\theta_m) \frac{di_{s\pm}}{dt} + i_{s\pm} \left( \frac{dL_{s\pm}(\theta_m)}{d\theta_m} \right) m_p \Omega_m \\ T_e = \frac{m_p}{2} K_c i_d^2 \end{cases}$$



- as usual IF you can compensate well the  $e$ ... Reg( $i_d$ ) is designed on and good P.S, sensor speed..

$$\frac{1}{R_s + sL(\theta_m)} \quad \text{T.F}$$

I need T.F changing with  $\theta_m$ .

I ORD T.F but Variable in  $\theta_m$ !

BUT the Best solution

is obtained with constant PI

controller on WORST

condition: for  $L_s(\theta_m)$

@ max value! (slower system)

sometimes solution

obtained by

ADAPTIVE CONTROL



$$\tau(\theta_m) = \frac{L_s(\theta_m)}{R_s}$$

time constant of load variable..

{ faster when you reach  $L_{min}$   
slower on  $L_{smax}$ !

↓  
attention to this

• there are non linearities ( $\sqrt{\cdot}$  and  $\cdot^2$ )  $\rightarrow$  { But if you compensate well, you predict the effect, OBTAIN ROBUST solution }

• What about id? What is id?

how to get

Information on idUEAs  $\Rightarrow$  to get this you need to measure

$i_{s1}, i_{s2}, i_{s3}, \dots, i_{sq}$  (depends on # phases)

$q$  sensors for the current needed...

NOT approached

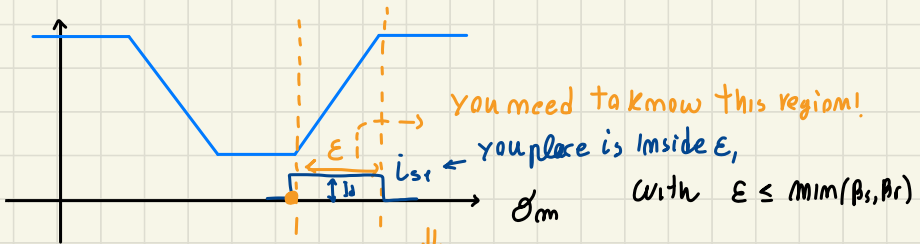
by space phasor theory!  $\rightarrow$  simple phase approach:

so other than  $i_{sJ}$   $J=1, \dots, q$ , to understand which is id I need additional information

$\hookrightarrow$  you need to know the angle  $\theta_m$

because:

from  $L_s(\theta_m)$



so you need to know the initial position  $\theta_m$  when switch on is

- how many steps you have in a turn with 2 poles machine...

# of sub period inside complete  $2\pi$  rotor turn?

$\hookrightarrow$  because this  $L_s(\theta_m)$  period is

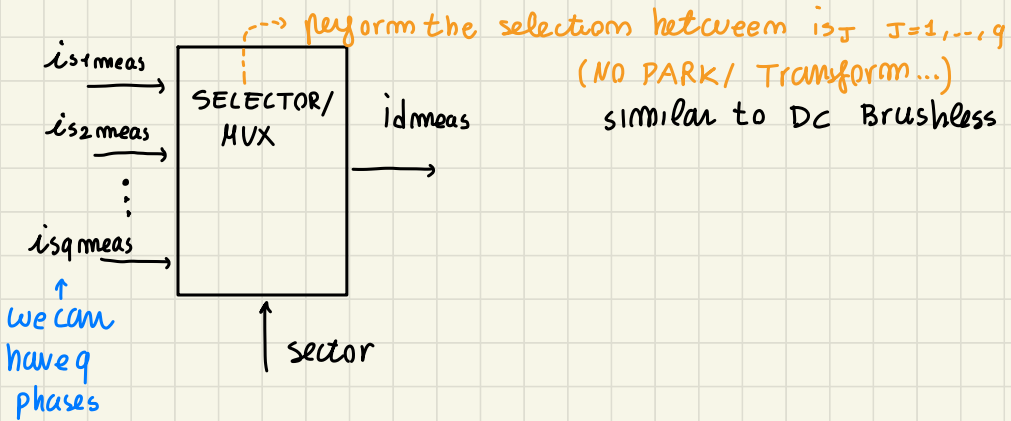
$\frac{2\pi}{N_r}$  with  $q$  steps of  
Nr current (# phases)  
& sub period

$\rightarrow$  overall you have  $N_r \cdot q$   
total current steps

$\rightarrow$  so the # of subsector is given by:

sector period =  $\frac{2\pi}{N_r q}$   $\rightarrow$  # sectors =  $N_r \cdot q$  (om om case +2)  
(Number of steps)  
 $\downarrow$   
on DC brushless we had 6...

- during  $2\pi$  period it switches 12 times  $\rightarrow$  you need to change the supplied coil 12 times  
(higher than in DC brushless)
- some way you need to have  $i_d$  meas as output of a SELECTOR/MUX



In order to maintain  $i_{s2} = I_d$  (you need  $I_d$  to have  $T_{emax}$ )

$$V_{s2} = \underbrace{R_s i_{s2}}_{u_d} + L_{s1}(\theta_m) \frac{di_{s1}}{dt} + \underbrace{i_{s2}}_{const} \underbrace{\frac{dL_{s1}(\theta_m)}{d\theta_m}}_{K_c \text{ constant}} \omega_m$$

$$T_e = \frac{m_p}{2} K_c i_d^2$$

emf proportional to the

speed  $\omega_m! \rightarrow$  so probably there is

probably you can  
define a value of  $\omega_m = \omega_b$

(base speed, such that you

reach max  $V_s$  you can apply by P.S)

$\leftarrow$  a value of  $\omega_m$  for which  
you reach  $V_{smax}$  (Base speed  $\omega_b$ )

## • BASE SPEED

$\omega_b$  to find it: when you have  $\rightarrow$  reached max VOLTAGE

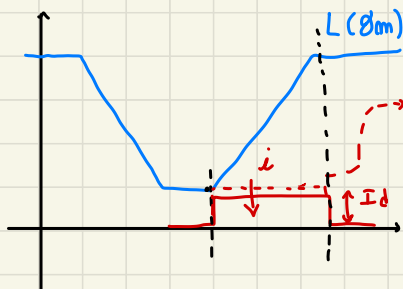
$$V_{dmax} = \underbrace{R_s I_d}_{\substack{\uparrow \\ \text{max voltage} \\ \text{by P.S}}} + \underbrace{I_d}_{\substack{\uparrow \\ \text{max current,} \\ \text{steady state}}} + I_d m_p K_c \omega_b$$

$$\Rightarrow I_d = \frac{V_{dmax}}{R_s + m_p K_c \omega_b}$$

$\uparrow$   
max current at  $\omega_b$

$$\omega_b = \frac{V_{dmax} - R_s I_d}{I_d m_p K_c}$$

IF you try to have  $\Omega_m > \Omega_b$  with same idea to maintain  $i_d = I_d$  square wave



go above  $\Omega_b$  with same strategy?

you have to reduce  $I_d$  ↓

in fact

$$\Omega_b = \frac{V_{dmax} - \overset{\text{negligible}}{R_s I_d'}}{m_p I_d' K_c} \quad \uparrow \text{ for } I_d \downarrow$$

you can reach

$$\Omega_m > \Omega_b \quad \Omega_m = \frac{V_{dmax} - R_s I_d'}{m_p I_d' K_c} \quad (I_d' < I_d)$$

$$I_d' = \frac{V_{dmax}}{R_s + m_p K_c \Omega_m} \sim \frac{1}{\Omega_m} \quad \begin{array}{l} \text{current} \\ \text{decreases} \\ \text{like } 1/\Omega_m \end{array}$$

negligible

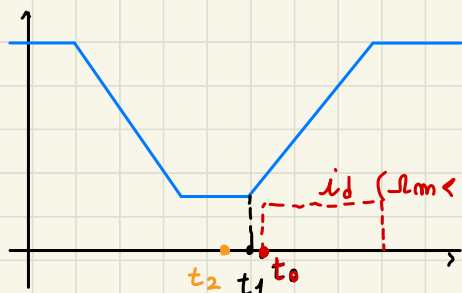
While the  
Torque ↓

$$T_e = \frac{m_p}{2} K_c (I_d')^2 \sim \frac{1}{\Omega_m^2}$$

(so here that's NOT  
a good approach...)

Torque decreases a  
lot for  $\Omega_m > \Omega_b$ !  
(more than the  
other elact drives, where  
om flux weakening  $T_e \sim 1/\Omega_m$ )

[use a similar approach of the one used for DC Brushless]



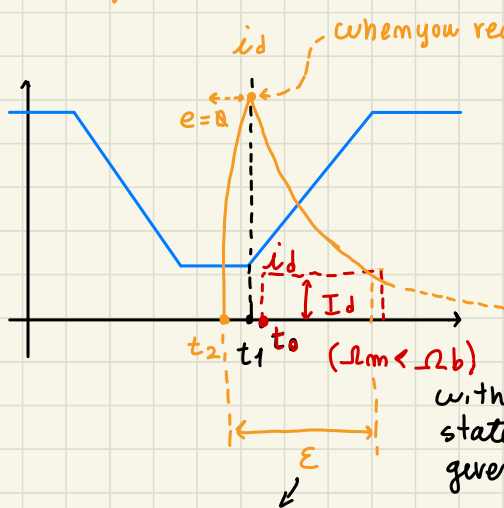
$t_1$ : point where  $L(\Omega_m)$  start  
increasing (pos. slope)

$i_d$  ( $\Omega_m < \Omega_b$ ) → simply a square wave

IF you start to supply the coil  
at time  $t_2 < t_1$  (he for  $dL/d\theta > 0$ )  
(not so much before: @  $t_2 \approx 0$ )

@  $t_2$ :  $e = 0$  because  $dl_s/d\omega_m = 0$  there is no emf

(different from DC brushless with trapezoidal waveform of emf)  
 on DC brushless we had some emf  $\rightarrow$  we had  $> 0$  forcing term  
 while here without emf  $i_d$  start to increase very fast!



when you reach  $t_1$ , what happens to  $i_d$ ?  
 (I hope it decreases!)

now @  $t_1$ :

$$e = i_d K_{cmp} \Omega_m$$

$i_d \gg I_d$   $\Omega_m > \Omega_b$

$e \gg$  than usual emf:  
 so it reduces a lot  
 the forcing term on

with steady  $\leftarrow$   $i_d$  equation  
 state condition of  $i_d(t)$   
 given by...

BUT, after  $E$   
 from  $t_2$ , you  
 have to START  
 with another  
 current supplied!

(you don't reach  $i_d$  steady  
 state)  $\rightarrow$  you should have  
 room for all currents



you have a leading  
 angle (anticipate) respect  
 previous  $i_d(t)$  shape..

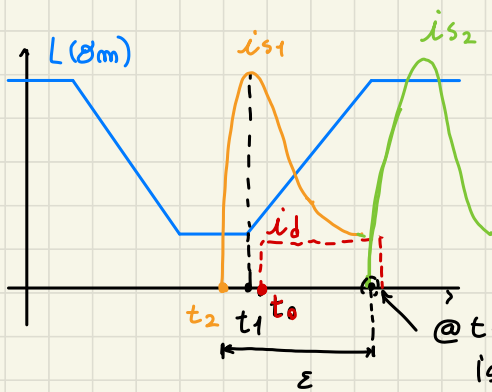
because  $e \sim i_d$  is also function of  $i_d$   
 which decreases, emf  $\downarrow$  the forcing term  $\downarrow$   
 there is a current  $i_d$  for which  $e = V_{dmax}$   
 so in steady state we reach

$$V_{dmax} = R_s i_d(+\infty) + 0 + i_d(+\infty) K_{cmp} \Omega_m$$

steady state  $\rightarrow$  positive current  $i_d \rightarrow 0$

$$i_d(+\infty) = \frac{V_{dmax}}{(R_s + K_{cmp} \Omega_m)}$$

some value of steady state as before  $> 0$   
 (inverse proportional to  $\Omega_m$ )



@  $t_2 + \epsilon$  is hard to STOP immediately  
 $i_{s2}(t)$ , you have inductance  
 (as 3phase or DC brushless)  
 $i_{s2}$  don't go to 0 immediately!  
 (some OVERLAPPING)

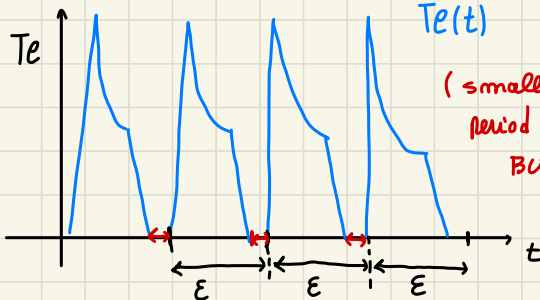
↓  
 Which effect has  
 on the TORQUE?

$$T_e = \frac{m_p}{2} k_c I_d^2$$

$k_c = \frac{dL}{d\theta}$  during  $(t_2, t_1)$   $k_c = 0 \rightarrow$  NO TORQUE!

so probably  $T_e$  has a shape like that:

(similar  
 to  $i_d$   
 shape but  
 squared)



(small  $\Delta t$  without any torque)  
 period during which  $i_d \neq 0$   
 BUT because  $k_c = 0$  you  
 HAVE NO TORQUE

• and for current limitation problems ↓

max current given by thermal problems...

(temperature of insulation) related to JOULE LOSSES ( $\sim$  RMS value of  $i_d$ )

pay attention to the fact that



The RMS value  
 of this  $i_d(t)$   
 must be limited

because of Joule losses  $\rightarrow$  Heat  $\rightarrow$  Temperature

(also service duty  
 and cooling syst)

- you need to start  $i_d$  on  $t_2 < t_1$  BUT NOT such that  $i_d$  reach a value too high

- another problem is related to current pick  $\rightarrow i_{smax}$  reached limited by P.S

limitation on  $i_s$  value

- $\left\{ \begin{array}{l} \text{RHS: to limit Joule losses} \\ \text{pick: to face P.S limitation, what switches can produce!} \end{array} \right.$

Now let's focus on the **POWER SUPPLY** of this machine:

very different from the other machines p.s seen...

use classical approach

with  $V_{dc}$  supply, and to control, we use to control  $i_d = I_d$  constant current (for  $\Omega_m < \Omega_b$ ) with a DC/DC converter

$\hookrightarrow$  consider a FULL BRIDGE for a single phase

that's 4 quadrant DC/DC converter (so you are able to produce)

$$\begin{cases} V_d \geq 0 \\ i_d \geq 0 \end{cases} \quad \text{drive } i_d \geq 0 \text{ inside coil}$$

and you can have 4 of this  
4 quadrant DC/DC...

(while on DC brushless you had 2 coils connected, Different!)

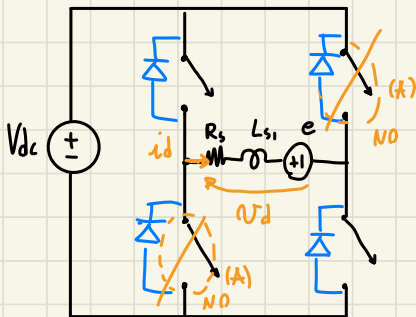
HERE you have 2 LEGS & coil

**BUT 4 quadrant is too much!** It's useful to have  $V_d \geq 0$

for when you need to control, reduce current by  $V_d < 0$

BUT I can have  $i_d < 0$  or  $i_d > 0$  NO need to control  $i_d$  sign  
( $T_e \propto i_d^2$ , not depend on sign( $i_d$ ))

4 switches in 2 LEGS are too much! We can remove 2 switches! (NO) (X)



reducing complexity by 2 DIODES and 2 Full switches...  
 considering  $i_d > 0$  only



• IF  $S1H$  and  $S2L$  closed: you apply a voltage  $V_d = V_{dc}$

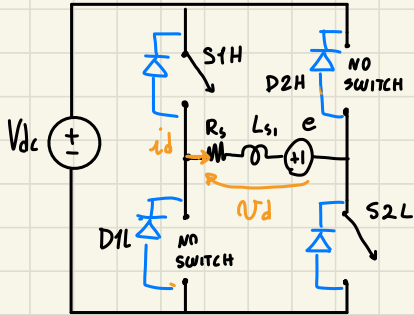
• IF  $S1H$  closed and  $S2L$  open: the current  $i_d$  move towards  $D2H$

so we apply  $V_d = 0$

• IF  $S1H$  open and  $S2L$  closed: current move towards  $D1L$

$V_d = 0$

• IF  $S1H$  and  $S2L$  open: current move through  $D2L$  and  $D2H$ , so you apply  $V_d = -V_{dc}$  to the load

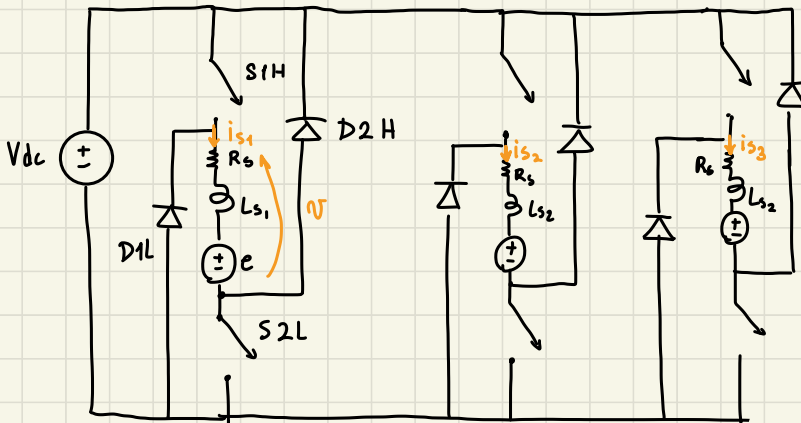


the Real P.S.  
 has a similar structure,  
 BUT:

usually drawn as:

OVERALL 3 phase scheme

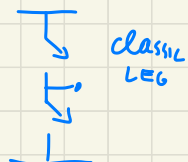
switch properly between the phases when  $\frac{dL}{d\theta} \neq 0$   $\begin{cases} > 0 \text{ Gen} \\ < 0 \text{ MT} \end{cases}$



6 switches  
 + DIODES (6)

2 quadrants DC/DC converter ( $V_d \geq 0$ )

(not easy to find on the market, different from classical LEG)



With this machine you can reach very HIGH speed!



used for VERY HIGH speed application (like on Dyson, washing machine)

---

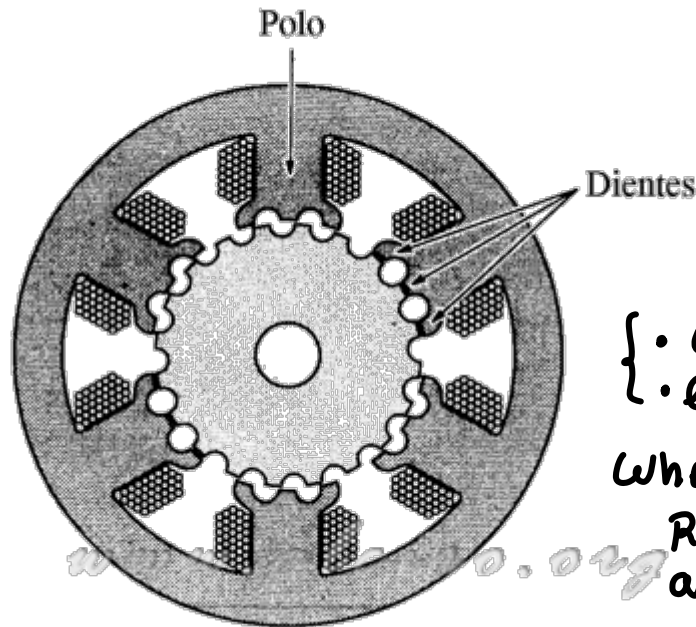
[NOT exam] DIFFERENT machine Based on the same equations

VARIABLE RELUCTANCE STEPPER MOTOR



# Variable Reluctance Stepper Motor

Hard to find on the market



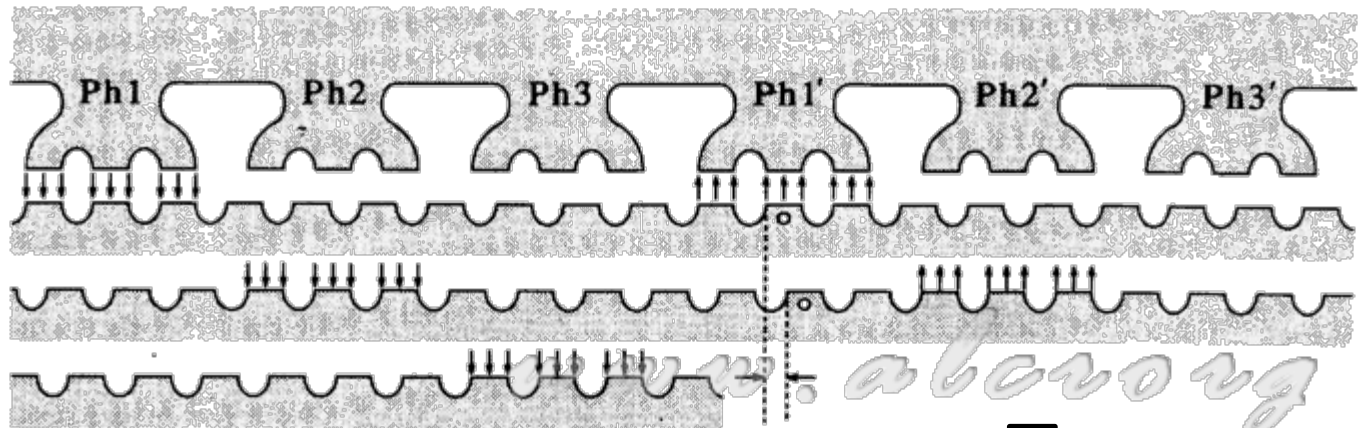
you have 6 terminals,  
similar to SRM

But with different ROTOR/STATOR  
pole number

- 6 stator poles with particular shape
- lots of poles!

When Tooth align you have alignment,  
Reluctance torque, and  $L_s(\theta_m)$  change with  
angle  $\theta_m$ !

according  
to how  
supply... →



usually controlled differently than SRM!

Angulo del paso = 

accordingly to how I supply it moves CW OR CCW

↓

after you change the supplied coil it moves by  $\frac{1}{3}$  by ROTOR distance  $\rightarrow \left(\frac{2\pi}{Nr}\right) = \text{distance between 2 Rotor poles}$   
 $q = \# \text{ phases}$

↓

$$\frac{2\pi}{qNr} \text{ overall (as } \epsilon)$$

But here you have lots of segments, small step!

↓

you can use upstep if you supply together differently...

sometimes you can divide  $\epsilon$  sub periods producing microsteps by same machinery

**stepper motor**  $\equiv$  usually used for LOW SPEED

↓ (while SRM for HIGH SPEED)

typically controlled without feedback, just give steps...

(as on ROBOT, you move ROTOR until you reach hope position)

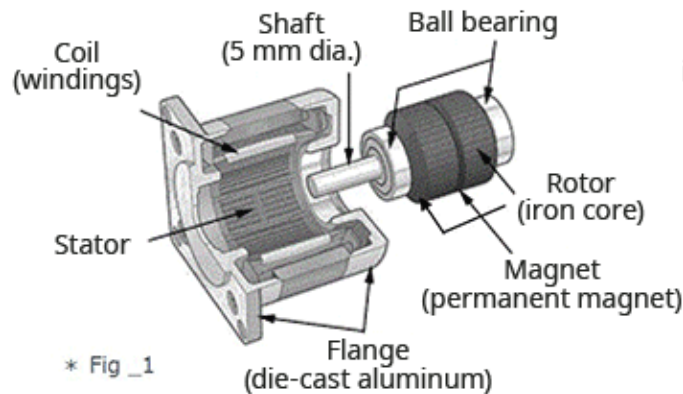
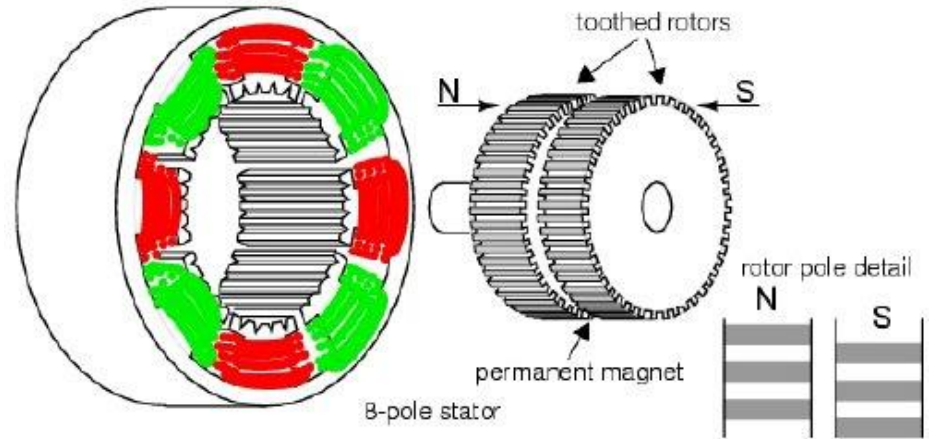
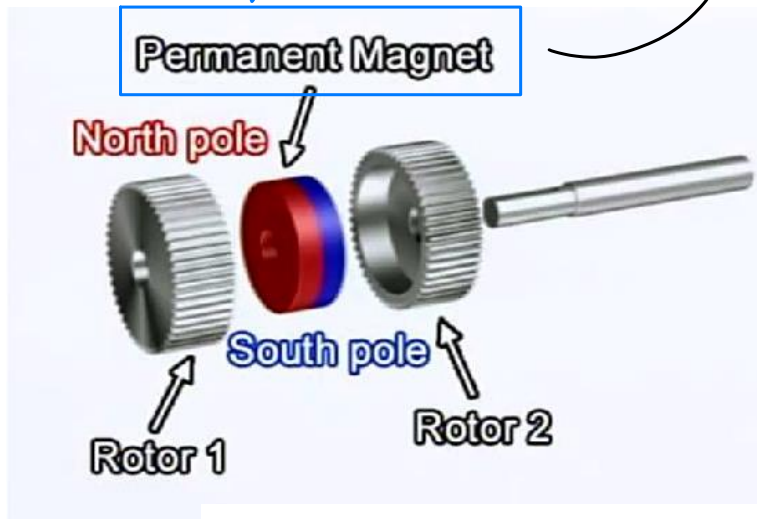
↓ to control this

you can use same eq of SRM!

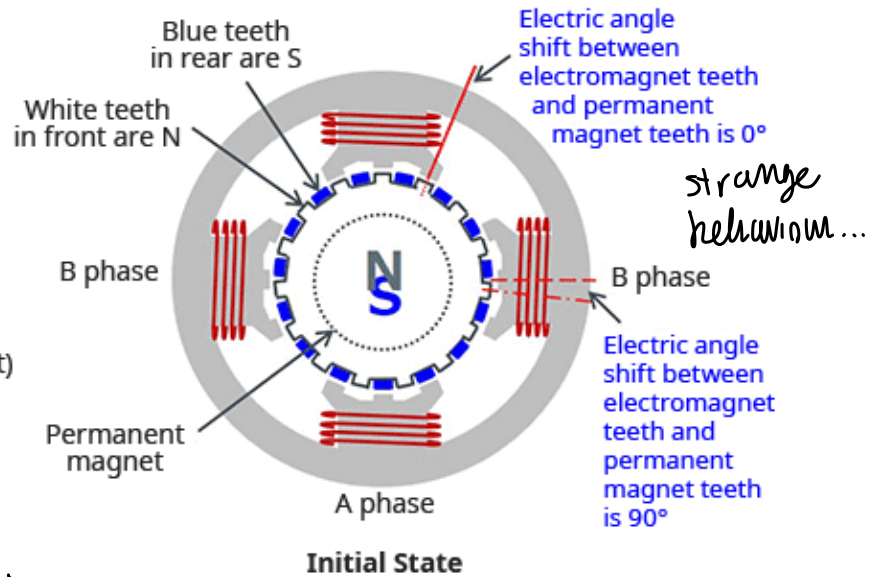
proximity sensor

# Hybrid Stepper Motor

create flux going radially to stator structure...  
(strange orthogonal flux lines..)



\* Fig \_1



**Initial State**  
In the initial state, currents are passed such that the upper pole of the A phase (upper/lower) is S and the lower pole is N

With only 2 phases, BUT careful on current sign! Torque given just by Reluct. torque BUT also excitation (due to PM)!

you can change current sign and NOTHING happens... supply to move by  $90^\circ$ ... move ROTOR in proper direction.. (CW, CCW)

↪ different from previous stepper MOTOR... While Here supply by 4 LEGS differently to have 4 quadrants DC/DC!

L<sub>D</sub>

Very HARD to model, simulate

DC brushless BUT here  $T_e = T_{REL} + T_{EXC}$ . .... Hand modelling

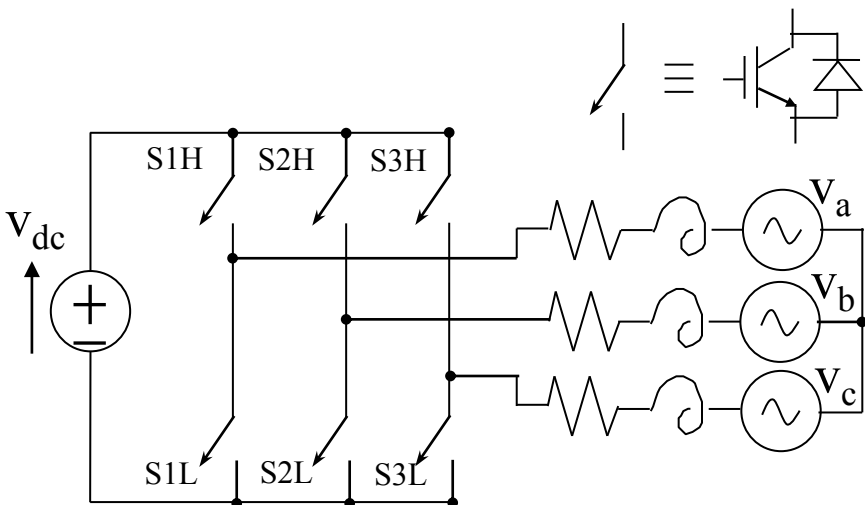
+ emf induced by PM!

used in TOYS, OR ROBOTICS for limited speed applications

for example previously used in ABB Robotic Arms!

used for LOW POWER / LOW SPEED

# DC/AC: three-phase voltage source inverter (VSI)



IDEA to use VSI to connect to 3 phase Grid...



OP. REGION of this inverter

Sinusoidal PWM with third harmonic injection (inscribed circle)

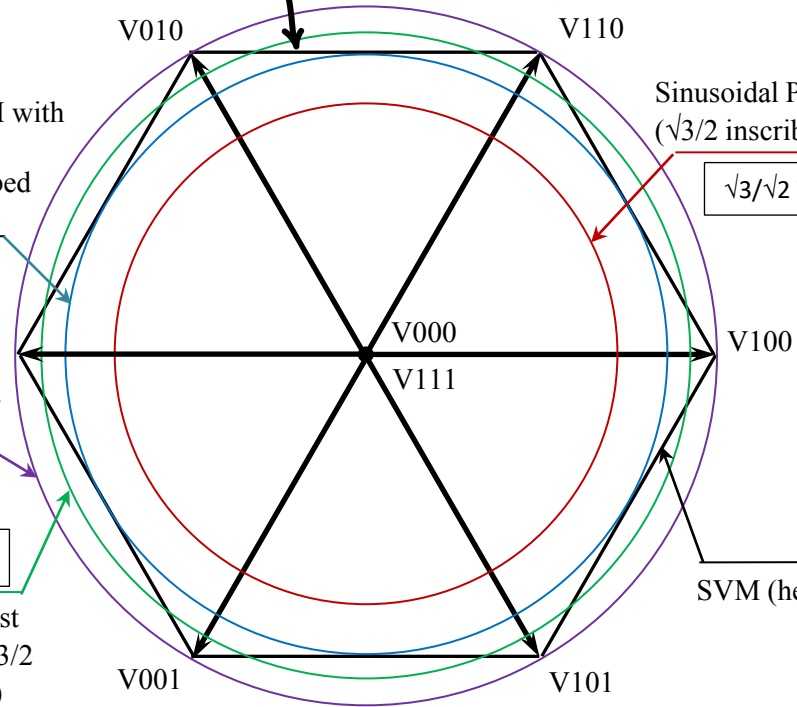
$1/\sqrt{2} V_{dc}$

Circumscribed circle

$\sqrt{2}/\sqrt{3} V_{dc}$

$2/\pi \sqrt{3}/\sqrt{2} V_{dc}$

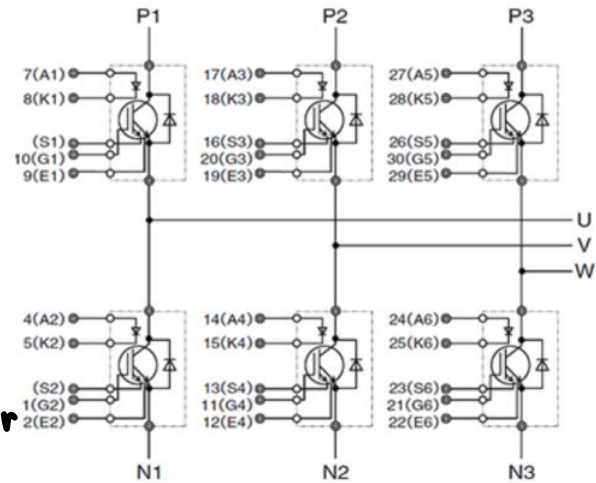
Square wave first harmonic ( $4/\pi \sqrt{3}/2$  inscribed circle)



Sinusoidal PWM ( $\sqrt{3}/2$  inscribed circle)

$\sqrt{3}/\sqrt{2} V_{dc}/2$

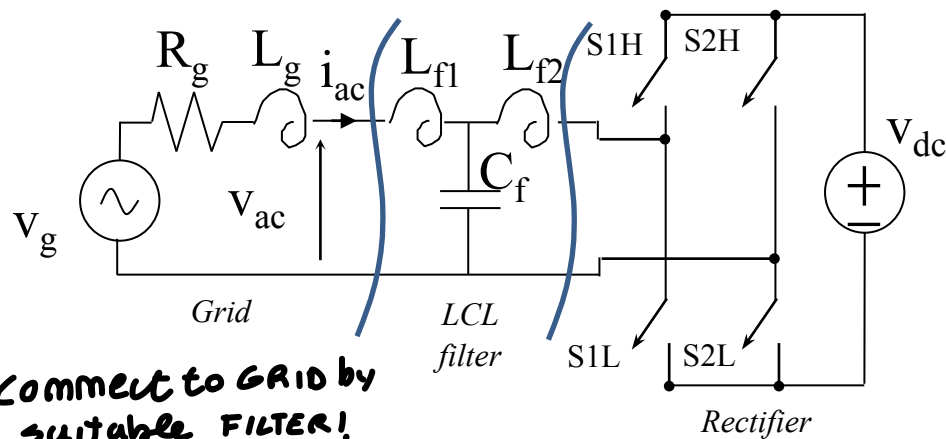
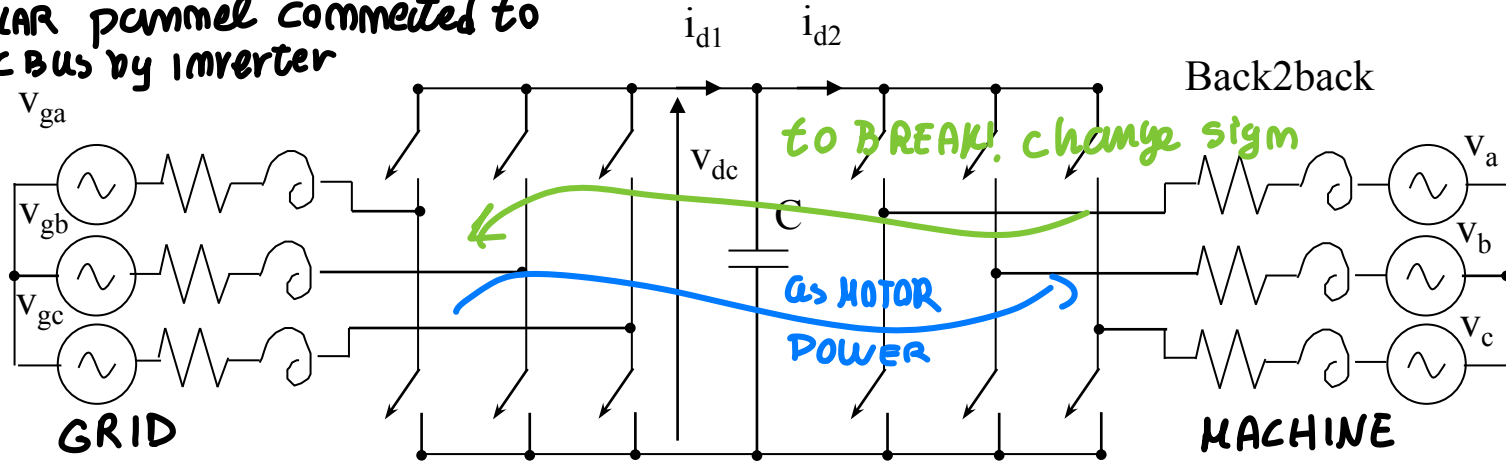
SVM (hexagon)



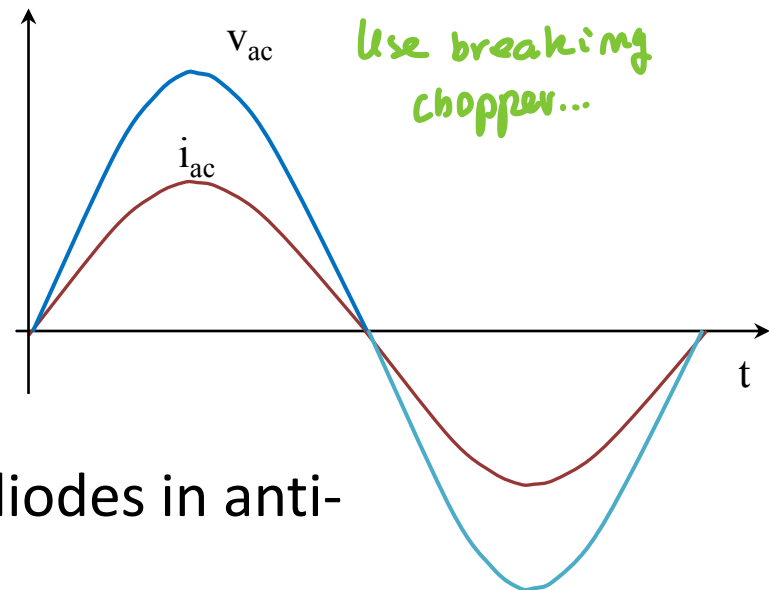
# Back-to-back configuration

synch mach, IM use inverter connected to the GRID!

Called also "Regenerative rectifier" or "Active Front End" (AFE)  
 SOLAR panel connected to DC BUS by inverter



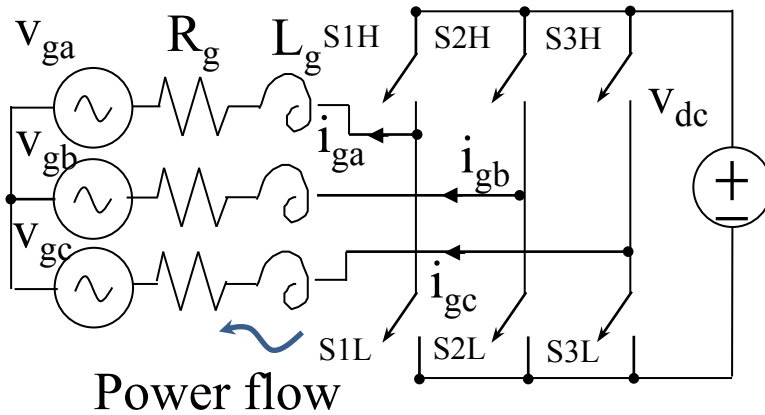
Connect to GRID by suitable FILTER!



Soft-starter is required (due to the diodes in anti-parallel to the switches)

# Regeneration

PROBLEM! by means of inverter you can produce 3 voltages  $GEM \rightarrow$  produce 3 controlled source



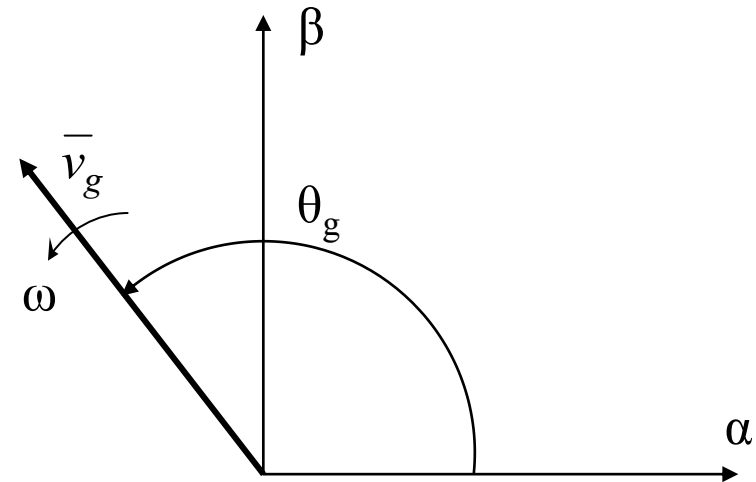
The grid has an equivalent circuit similar to that of an ac machine.

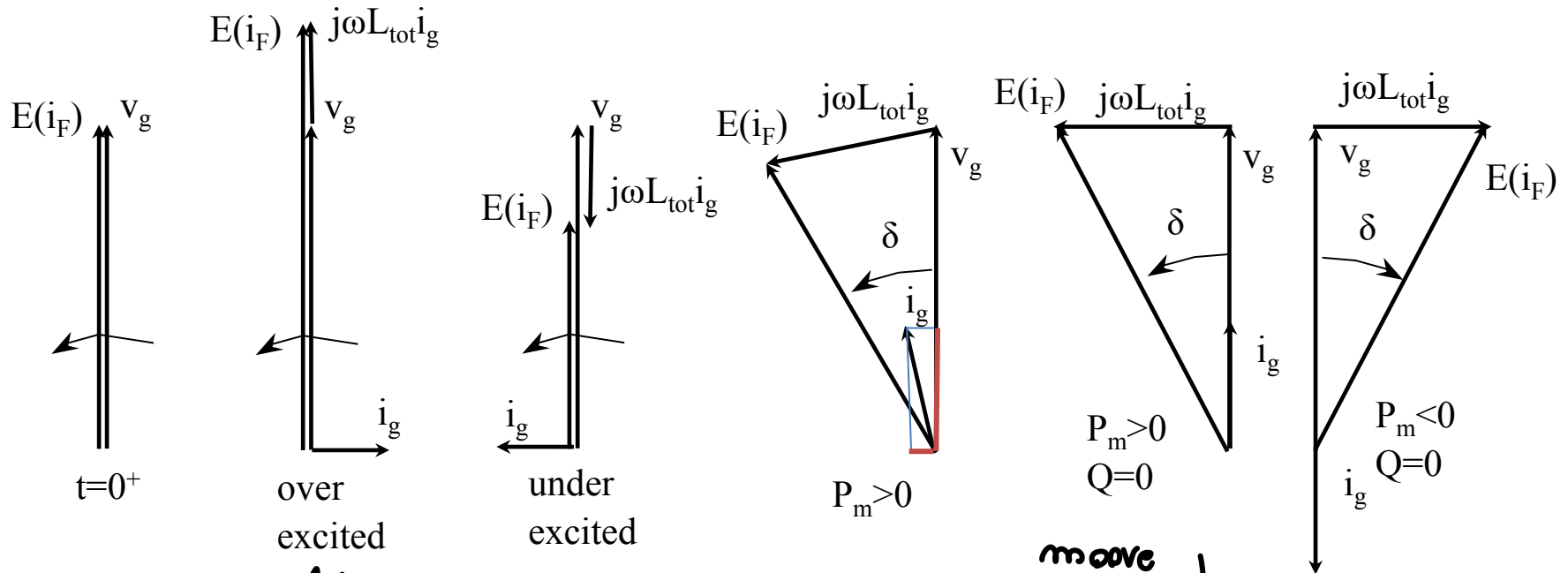
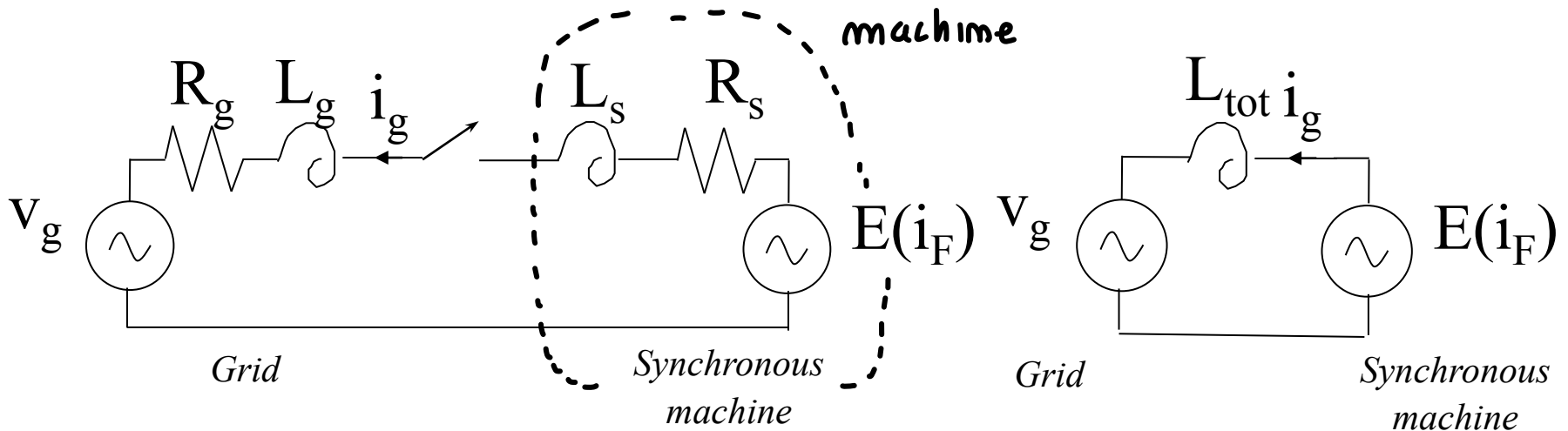
The main difference is that the voltage and the frequency are quite constant (in an ac machine the emf and the frequency are quite proportional to the mechanical speed)

The space phasor formula may be applied to the grid voltage  $v_g$ . In a stationary frame ( $\alpha\beta$ ), it results ( $v_{ga} = V_{ph\ max} \cos(\theta_g)$ ):

$$\overline{v_g} = V_g e^{j\theta_g}$$

Where  $V_g$  is the rms value of the line-line voltage, while  $\theta_g$  the angle between the space phasor  $v_g$  and the  $\alpha$ -axis





DRIVE ACTIVE/ REACTIVE power as I wish → energy!

# Control of power flow into the grid

Call  $v_i$  the voltage produced by the VSI (limited to the hexagon operating area). Choose a reference frame fixed with the grid voltage, where the imaginary axis (q) is aligned with the space phasor  $v_g$ . It results: (equation similar to AC Brushless)

$$\bar{v}_i = R_g \bar{i}_g + L_g p \bar{i}_g + j \dot{\theta} L_g \bar{i}_g + \bar{v}_g$$

The components along d and q axes are:

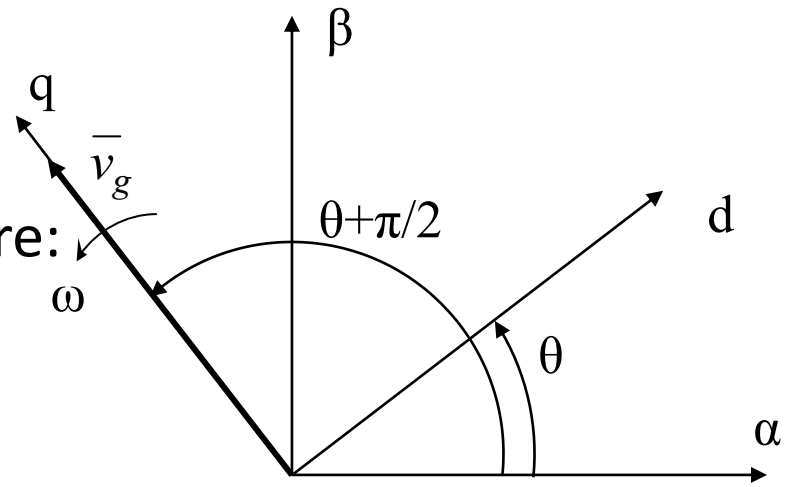
$$v_{id} = R_g i_{gd} + L_g p i_{gd} - \dot{\theta} L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + L_g p i_{gq} + \dot{\theta} L_g i_{gd} + v_g$$

The active and reactive power fed into the grid are:

$$\begin{cases} P_g = \text{Re}(\bar{v}_g \underline{i}_g) = v_g i_{gq} \\ Q_g = \text{Im}(\bar{v}_g \underline{i}_g) = v_g i_{gd} \end{cases}$$

I try to align with Grid  
Voltage := VOLTAGE ORIENTED control!  
(supply current in phase with  $\bar{v}_g$  := only P)  
 $\perp \bar{v}_g$  := only Q



# Control of active power fed into the grid

The derivative of the angle  $\theta$  is the frequency  $\omega$ .

Consider a steady state condition and a grid voltage constant in amplitude and frequency:  $\overline{v_g} = V_g e^{j\omega t}$ . Looking from the reference frame (dq), all the quantities are constant. Therefore:

$$\dot{\theta} = \omega$$

$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

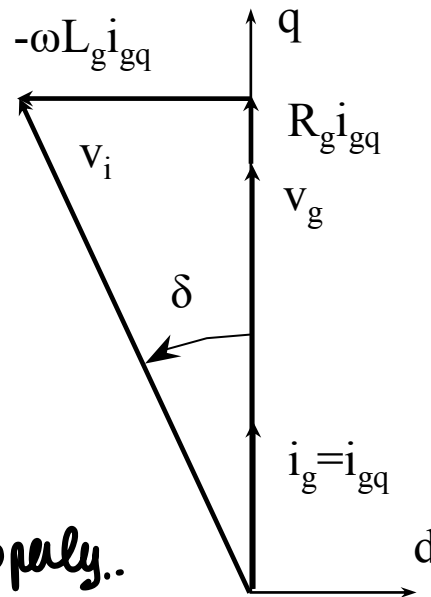
If the current  $i_{gd}$  (component along the d axis) is fixed to zero, the phasor diagram is:

$$v_{id} = -\omega L_g i_{gq}$$

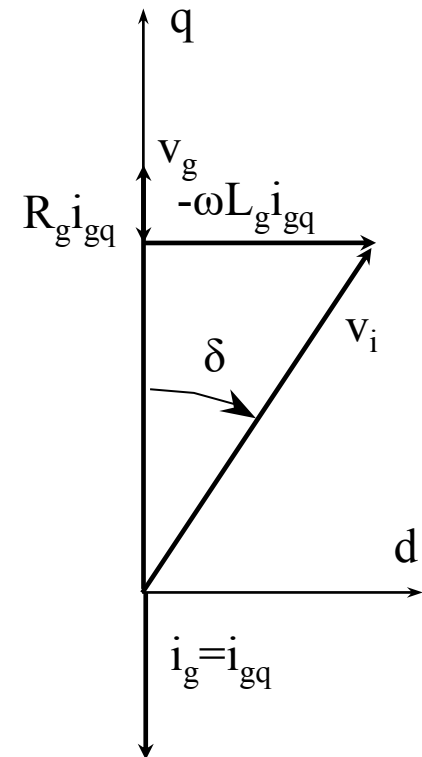
$$v_{iq} = R_g i_{gq} + V_g$$

*Drive the VOLTAGE properly..  
SIMILAR to DC Brushless..  
OR different  $P \geq 0$  sign*

Regenerative &  $Q_g=0$



Rectifier &  $Q_g=0$



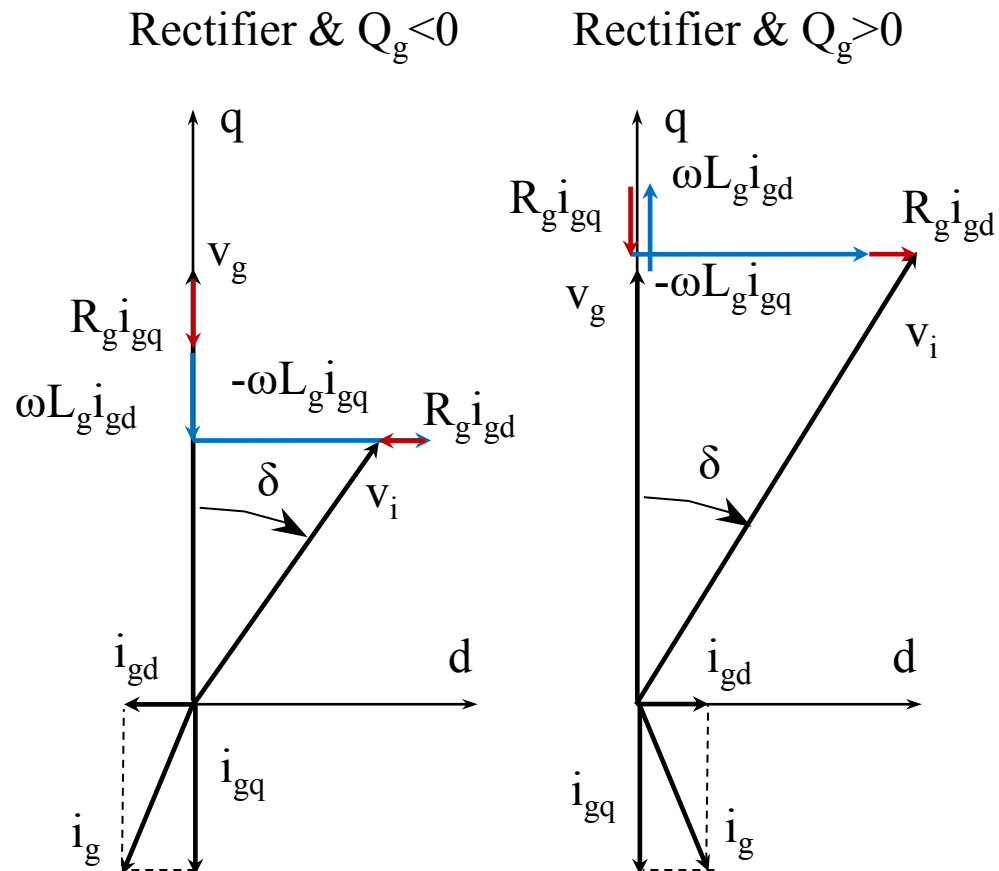
# Control of active/reactive power fed into the grid

Controlling the current  $i_{gd}$ , it is possible to drive the reactive power: a positive value means that the inverter looks like a capacitor (and the grid like an inductor); vice versa for a negative value.

$$v_{id} = R_g i_{gd} - \omega L_g i_{gq}$$

$$v_{iq} = R_g i_{gq} + \omega L_g i_{gd} + V_g$$

*The figure represents the conditions when the active power is negative (operation as rectifier). Similar approach may be done for positive value of the active power (regeneration)*

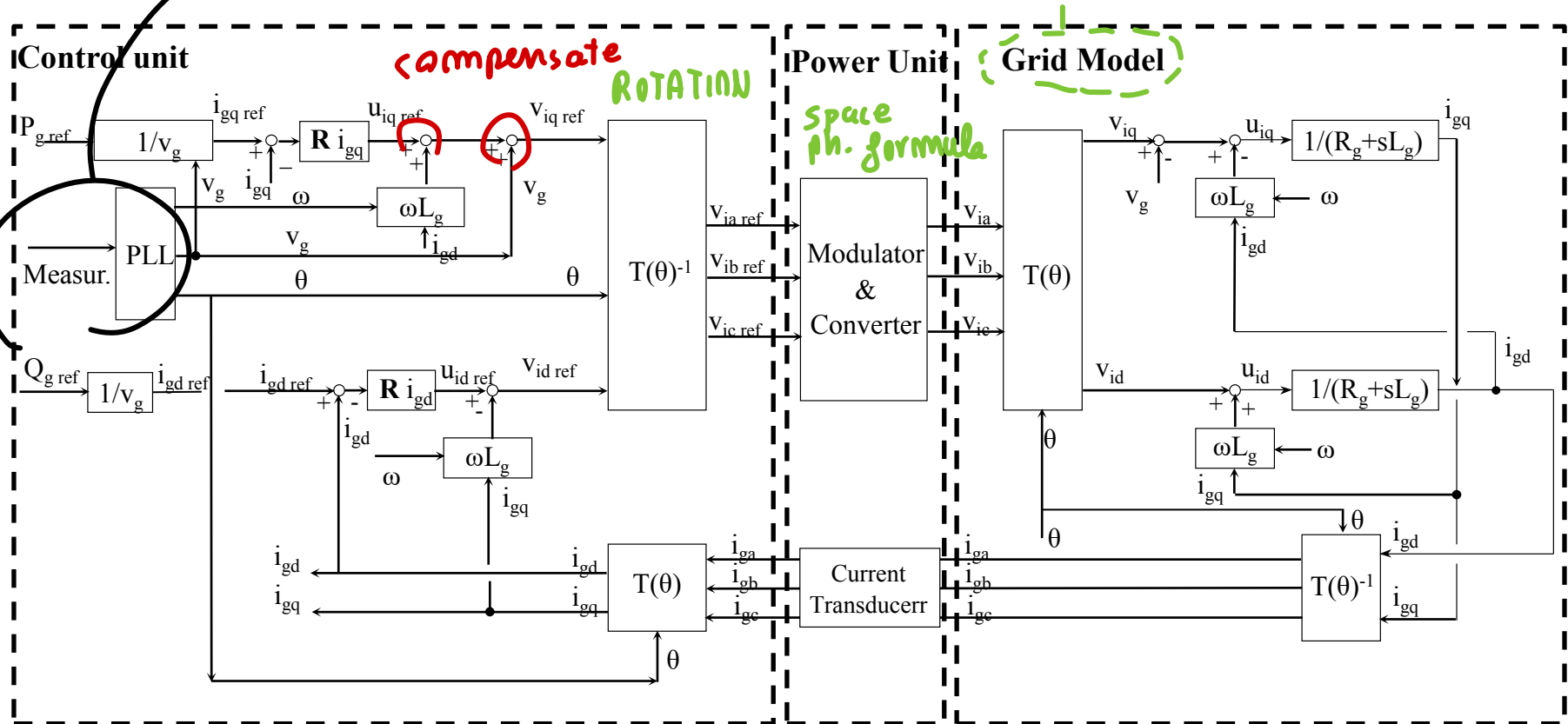


# Control schemes

IF you know where is  $\bar{V}_g$ ,  
produce Ref fixed with  $\bar{V}_g \rightarrow V_{OC}$

to know where is GRID  
value, to compensate drop

similar to AC Brushless

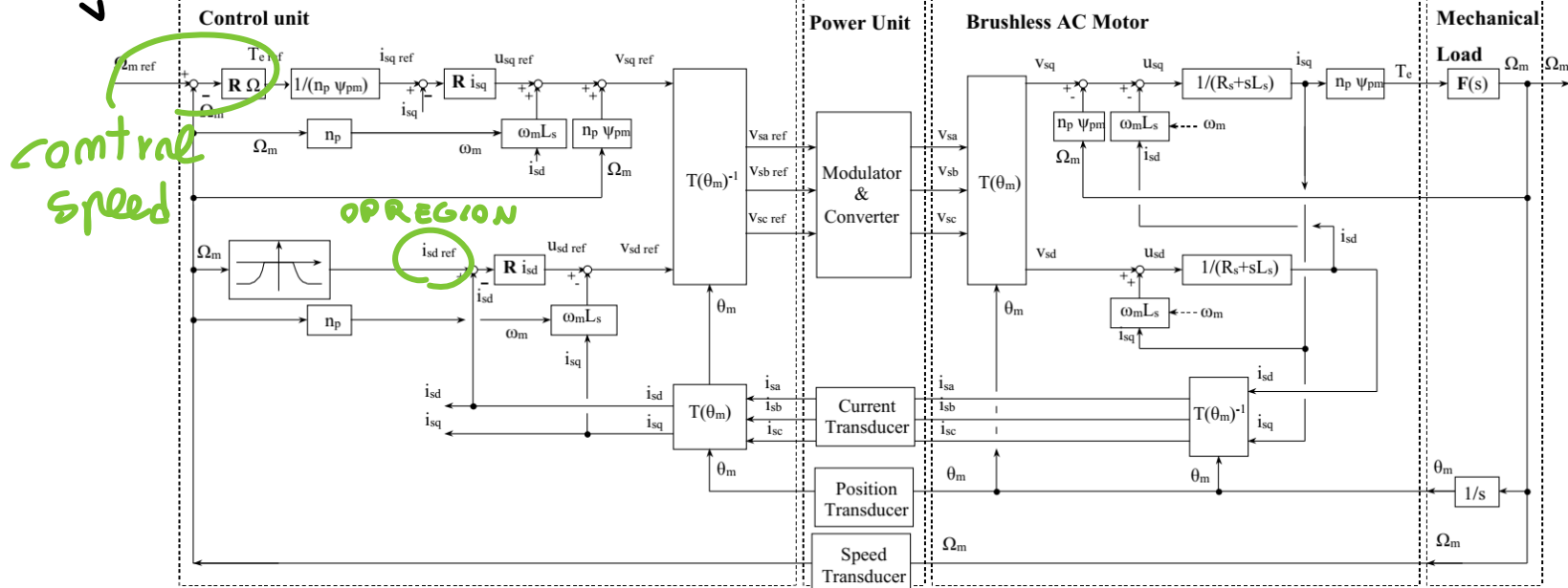
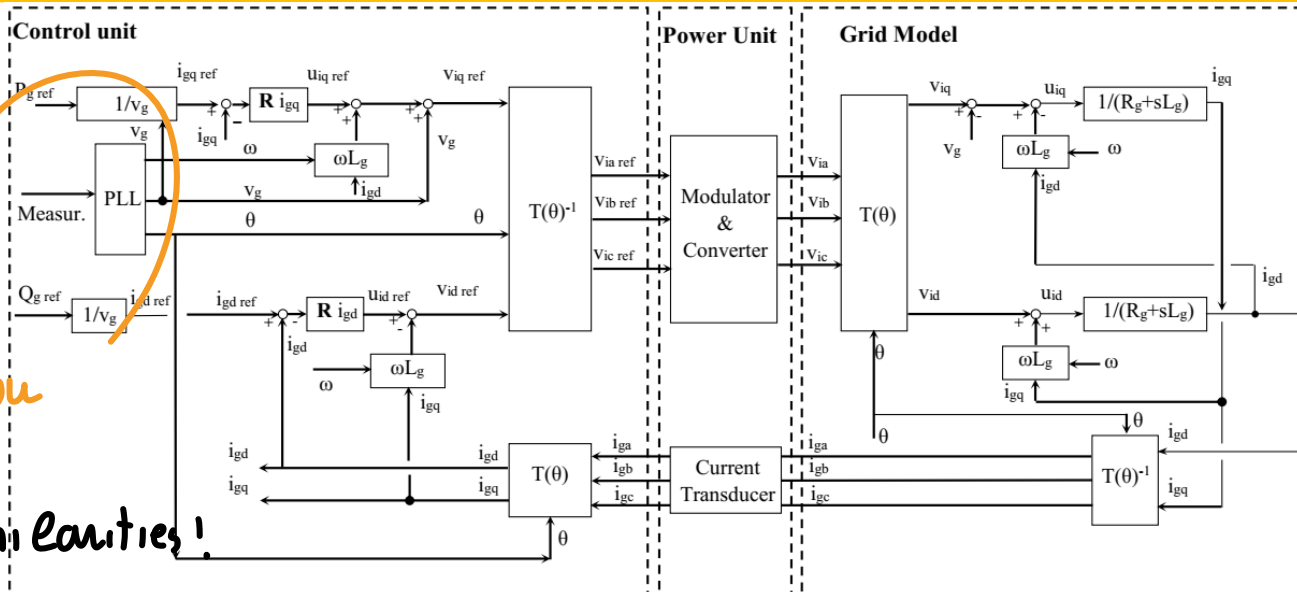


# Control schemes: comparison with ac brushless

On GRID  
you give  
 $P_g, Q_g$   
REF by  
request!  
NOT GO as you  
WANT!

Similarities!

In base of  
request:  
(rephase...)  
INVERTER  
work as  
capacitor..  
Sometimes  
STOP IF  
needed...



manage GRID problems, short circuit ...



IDEA to continue

to supply, small power →

introduce current  
for safety open  
switches!

You can use FOC to control power inside the Grid



VOC "VOLTAGE ORIENTED"  
CONTROL